COMM061 Natural Language Processing

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Introduction

The widespread deployment of NLP systems across culturally and linguistically diverse communities has revealed a systematic failure mode: models trained on standard, inner-circle English varieties degrade substantially when confronted with dialectal variation [1], [2]. This is particularly consequential for pragmatic tasks such as sarcasm detection, where the surface form of an utterance is deliberately incongruent with its intended meaning [3], a mismatch that is realised through variety-specific conventions including Australian hyperbolic understatement, British dry wit [2], and Indian English code-switching patterns that blend Hindi and English lexical items [1].

We address this challenge within the context of the BESSTIE dataset [1], which spans British (en-UK), Australian (en-AU), and Indian (en-IN) English across multiple domains. The goal is to analyse dataset characteristics and class imbalances, evaluate model performance across varieties through controlled experimental setups (including baselines, pre-trained language models [4], [5], and LLM-based approaches [6], [7], [8]), and ultimately improve robustness and cross-variety generalisation in dialect-aware sarcasm detection systems.

1 Dataset Analysis (Visualisation and Vocabulary Analysis)

1.1 Label Distributions and Class Imbalance

The BESSTIE dataset [1] contains 6,243 instances in total, distributed across train (3,747), validation (313), and test (2,183) splits, as summarized in Table 1. Each instance belongs to one of three English varieties, Australian (en-AU), Indian (en-IN), and British (en-UK), and is labelled for both Sentiment and Sarcasm as binary classification tasks.

Split	Rows	en-AU	en-IN	en-UK	Google	Reddit	Positive	Negative	Sarcastic	Not Sarcastic
Train	3747	1145	1399	1203	1874	1873	1840	1907	524	3223
Validation	313	95	117	101	167	146	153	160	44	269
Test	2183	667	816	700	1101	1082	1066	1117	305	1878

Table 1: Dataset Summary across Train, Validation and Test Splits.

Sentiment Distribution. As shown in Figure 1, sentiment labels are broadly balanced across all three varieties. en-AU is mildly skewed towards negative sentiment (54% Negative, 46% Positive), while en-IN (50.7% / 49.3%) and en-UK (48.5% / 51.5%) sit close to an even split. This near balance suggests that sentiment classification should not suffer significantly from label imbalance, and standard training without oversampling is likely to be adequate for this task.

Sarcasm Distribution. Sarcasm tells a very different story. en-AU is the least imbalanced of the three, yet it is still heavily skewed at 70.6% Not Sarcastic versus 29.4% Sarcastic. en-IN and en-UK are far more extreme, with sarcastic instances making up only 6.8% and 7.6% of their totals respectively. This level of imbalance creates a serious risk in model training, as shown in [9], class imbalance in sarcasm detection tasks significantly affects supervised classification performance, and a naive classifier that always predicts "Not Sarcastic" would achieve over 90% accuracy on en-IN and en-UK while being completely useless. For this reason, class imbalance must be addressed through oversampling, class weighting, or the use of Macro-F1 as the primary evaluation metric before any meaningful comparison across models can be made.

Split Consistency. Figure 2 confirms that the class imbalance is consistent across the Train, Validation, and Test splits for all varieties. This rules out the possibility that skewed results might arise from an unrepresentative split and confirms that the dataset design is internally coherent.

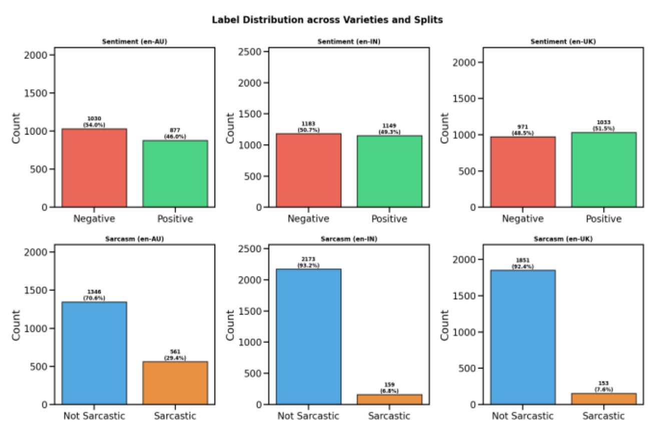

Figure 1: Label Distribution across Varieties and Splits.

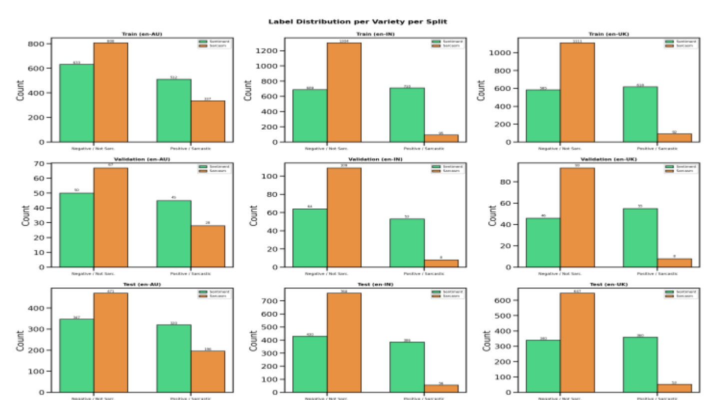

Figure 2: Label Distribution per Variety per Split.

Domain Effects. The source domain introduces an additional layer of complexity that interacts strongly with the sarcasm task. As shown in Figure 3, sarcasm is essentially absent from Google Places reviews across all varieties, with rates of 5.7% for en-AU, 1.0% for en-IN, and just 0.1% for en-UK. Reddit is the primary source of sarcastic content, with en-AU showing the highest rate at 41.7%, followed by en-UK at 21.0% and en-IN at 12.9% as illustrated in Figure 4. For sentiment, Google reviews are overwhelmingly positive across all varieties (around 74-75%), while Reddit skews negative. This effect is most pronounced in en-UK, where only 11% of Reddit posts carry positive sentiment. The practical consequence of this is that a model trained on Google reviews will encounter a very different linguistic register and label distribution compared to one trained on Reddit. This domain gap has to be considered when interpreting cross-variety results, as it can easily be mistaken for a variety gap [10].

POS Distribution. Beyond label counts, Figure 5 reveals meaningful structural differences in how the three varieties use language. Nouns are the dominant part of speech in all varieties, but en-IN shows notably higher noun usage at around 19.8%, compared to approximately 18.1% for en-AU and 18.5% for en-UK. More importantly, en-IN uses fewer determiners (around 6.5% vs roughly 8% for both inner-circle varieties) and shows noticeably lower proper noun usage as well. Lower determiner frequency in en-IN is consistent with well-documented features of Indian English grammar, where articles such as "a" and "the" are commonly omitted. This reflects genuine grammatical divergence between the varieties rather than just surface-level vocabulary differences, and has direct implications for how well models trained on inner-circle varieties might transfer to en-IN.

1.2 Vocabulary Analysis

Word Frequency Distributions. The top-20 word frequency plots in Figure 7 show a shared core vocabulary across all three varieties, with words like "good", "food", "place", and "service" dominating non-sarcastic content. This reflects the Google Places review domain and suggests that the varieties converge on similar topics in everyday writing. However, variety-specific signals emerge clearly when looking at sarcastic texts in Figure 6. en-AU sarcasm frequently features words like "government", "labor", and "australia", pointing to a thread of political discourse in Australian Reddit communities. en-UK mirrors this pattern with "labour" and "country" appearing prominently. The most distinctive finding comes from en-IN, where sarcastic texts contain tokens such as "hai", "toh", "mai", and "kya", all of which are Hindi words mixed into otherwise English sentences. This provides direct evidence of Hindi-English code-mixing in the Indian English subset [11], and these tokens would be completely out-of-vocabulary for standard English pre-trained models.

Vocabulary Overlap - Jaccard Similarity. The Jaccard similarity matrix in Figure 8 measures raw token overlap between varieties. en-AU and en-UK share the most vocabulary at 0.2576, which is consistent with both being inner-circle varieties with similar cultural and topical contexts. Both pairings with en-IN score lower, at 0.2177 for en-AU vs en-IN and 0.2244 for en-UK vs en-IN. While the differences between these scores are modest, the pattern is consistent: the outer-circle Indian variety is lexically more distant from both inner-circle varieties than they are from each other. All

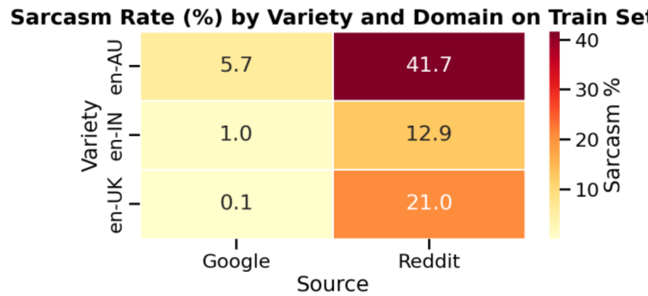

Figure 3: Sarcasm Rate (%) by Variety and Domain.

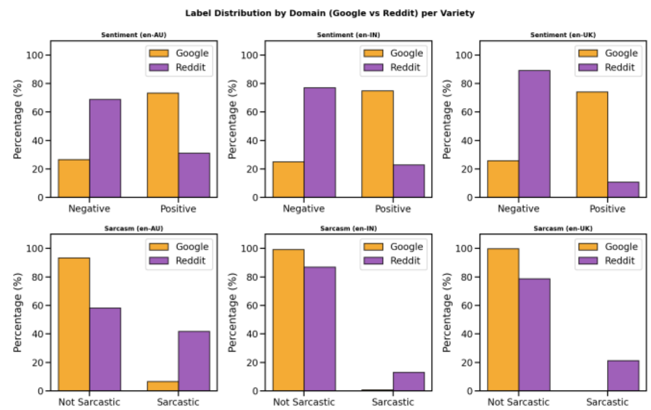

Figure 4: Label Distribution by Domain (Google vs Reddit) per Variety.

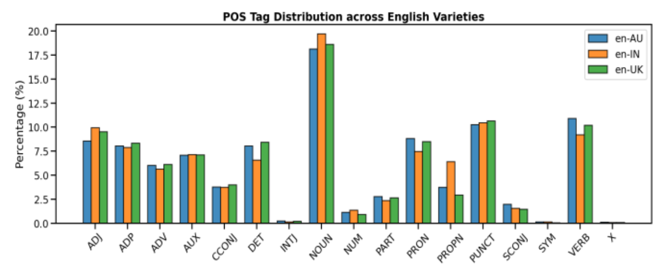

Figure 5: POS Tag Distribution across English Varieties.

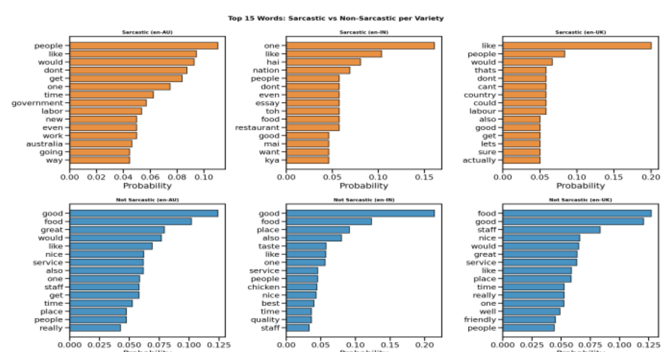

Figure 6: Top 15 words: Sarcastic vs Non-Sarcastic per Variety.

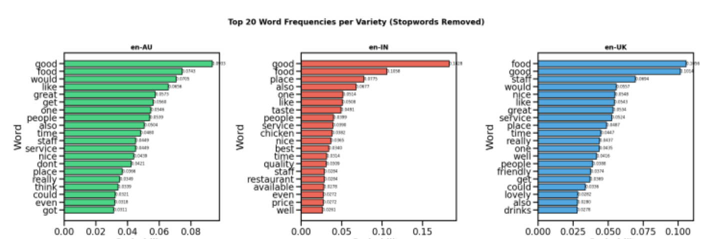

Figure 7: Top 20 words Frequencies per Variety (Stopwords Removed.

scores fall in the range 0.21 to 0.26, which is low in absolute terms, reflecting a large proportion of variety-specific vocabulary across the board.

Vocabulary Overlap - TF-IDF Cosine Similarity. Cosine similarity over TF-IDF representations in Figure 9 captures a different angle: not just whether two varieties share words, but whether the important words are shared. The results shift noticeably here. en-AU and en-UK score 0.8649, suggesting that the most semantically important terms are highly shared between these two inner-circle varieties. en-UK vs en-IN scores 0.7954 and en-AU vs en-IN scores 0.7357, both of which are lower but still reasonably high. This gap is consistent with the Jaccard results, though cosine similarity reveals that en-IN does share a substantial portion of high-weight vocabulary with the inner-circle varieties. Taken together, the two measures paint a nuanced picture: the varieties differ in their full vocabulary profiles but still share a meaningful core of important terms.

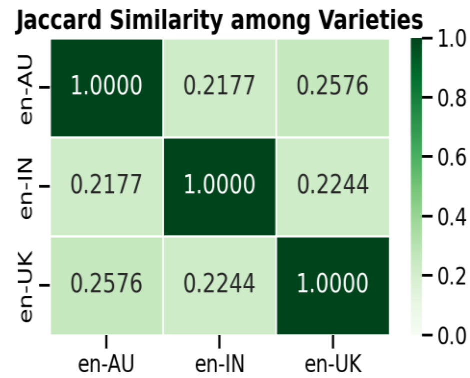

Figure 8: Jaccard Similarity among Varieties.

Linguistic Distance. Linguistic distance refers to the degree to which two language varieties diverge in vocabulary, grammar, and pragmatic conventions. When the distance between a training variety and a test variety is high, a model has to generalise across different linguistic systems rather than just different topics, which is a harder task. The evidence gathered here suggests that the distance between en-IN and the inner-circle varieties is real but partial. The Jaccard gap is modest in magnitude, and much of the raw vocabulary difference is likely driven by topical divergence. For example, Indian reviews frequently mention food terms like "chicken" and "taste" that reflect local cuisine, while Australian sarcasm is shaped by local political events. However, the POS analysis provides stronger evidence of genuine structural divergence. The lower determiner frequency and higher noun density in en-IN are consistent with documented grammatical features of Indian English that go beyond topic differences [3]. The presence of Hindi code-mixed tokens such as "hai" and "kya" in sarcastic en-IN texts [11] further confirms that this variety distance is partly linguistic in nature. We therefore expect that models fine-tuned on inner-circle varieties will face a non-trivial challenge when applied to en-IN, particularly for the sarcasm task where cultural and pragmatic context matters more than in sentiment classification.

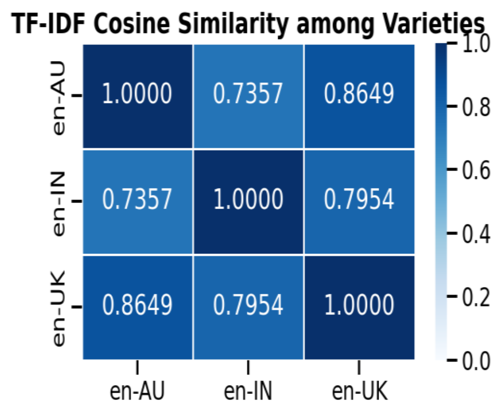

Figure 9: TF-IDF Cosine Similarity among Varieties.

1.2.1 Synthetic Data Comparison

To investigate whether a synthetically generated dataset could supplement the real BESSTIE corpus [1], we produced 1,244 LLM-generated instances using GPT-5 [25] following the same structure as BESSTIE, with text, variety, source, Sentiment, and Sarcasm labels across all three English varieties. We then compared this synthetic corpus against a size-matched BESSTIE sample using the same vocabulary and distributional analysis applied in Sections 1.1 and 1.2.

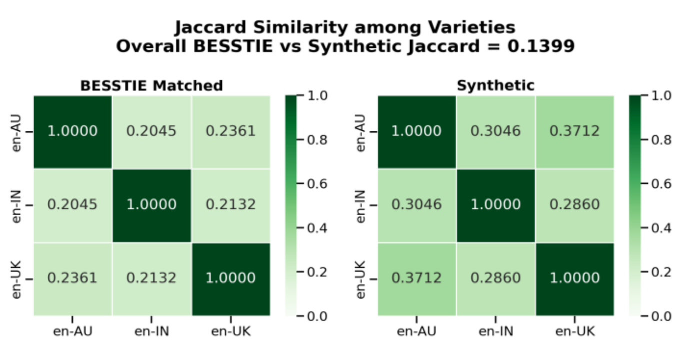

Figure 10: Jaccard Similarity: BESSTIE vs Synthetic.

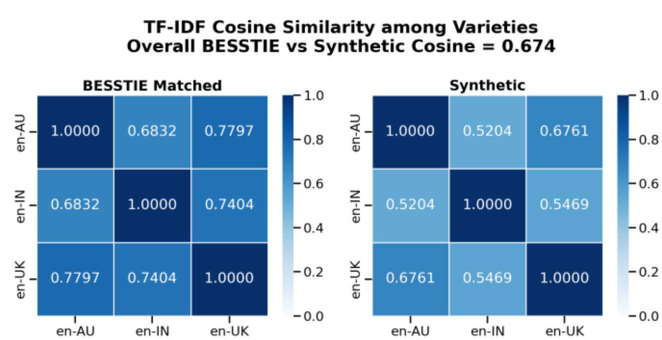

Figure 11: TF-IDF Cosine Similarity: BESSTIE vs Synthetic.

The overall Jaccard similarity between the two corpora was 0.1399 and the TF-IDF cosine similarity was 0.674, as shown in Figure 10 and 11. While some surface-level vocabulary is shared, the two datasets diverge considerably in their lexical content. The BESSTIE vocabulary contained 9,429 tokens not found in the synthetic data at all, compared to just 830 tokens unique to the synthetic set. This asymmetry shows that the synthetic corpus covers a much narrower vocabulary range and misses a large portion of the dialectal richness found in the real data.

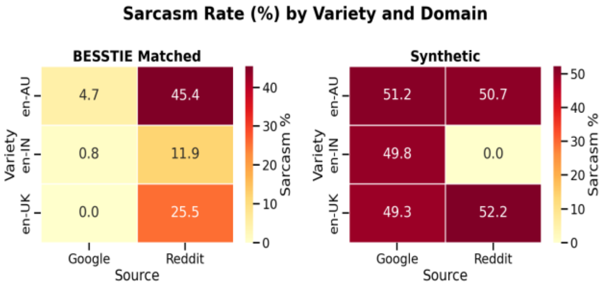

Figure 12: Sarcasm Rate (%) by Variety and Domain: BESSTIE vs Synthetic.

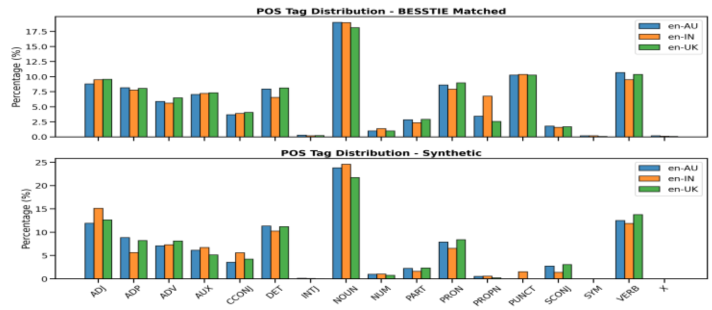

Figure 13: POS Tag Distribution across English Varieties: BESSTIE vs Synthetic.

The label distributions reveal an even more critical difference. As illustrated in Figure 12, the synthetic data has sarcasm rates of around 50% across almost all variety and domain combinations. This is completely at odds with the natural imbalance in BESSTIE [1], where sarcasm rates range from 0.0% in en-UK Google reviews to 45.4% in en-AU Reddit posts. The synthetic data essentially treats sarcasm as a coin flip, which does not reflect how sarcasm occurs in real language and would produce overly optimistic model performance if used for training or evaluation without correction [9].

The POS comparison in Figure 13 further highlights structural differences. Synthetic texts show noticeably higher noun usage (around 24% for both en-AU and en-IN) compared to roughly 18-19% in BESSTIE, along with much lower punctuation rates. This suggests the LLM-generated texts tend towards clean, well-formed sentences that lack the informal punctuation patterns typical of Reddit posts. Perhaps most importantly, the within-variety Jaccard scores in the synthetic data are consistently higher than in BESSTIE across all pairs, meaning the synthetic varieties look more similar to each other than their real counterparts do. This suggests the synthetic generation process partially flattens the variety gap that the coursework is specifically designed to study.The overall comparison metrics are summarized in Table 2. Based on these findings, the synthetic data was not used to augment model training. While it may be a useful tool for rapid prototyping, it does not faithfully reproduce the distributional properties, class imbalance, or linguistic diversity of the real BESSTIE corpus.

Metric	BESSTIE (matched)	Synthetic
Total rows	1244	1244
Vocabulary size	11097	2498
Average Word Length	49.19	10.33
Median Word Length	34	10
Sarcasm rate (%)	14.7	50.6

Table 2: BESSTIE vs Synthetic Overall Comparison Summary.

1.3 Class Imbalance Handling - Oversampling vs Class Weighting

Given the severe sarcasm imbalance identified in Section 1.1, we tested three training strategies using TF-IDF + Logistic Regression as a controlled baseline to determine which approach best handles the minority class before passing training data to downstream experiments. Results were averaged over 3 random seeds to account for variance [9]. As shown in Figure 14 and Table 3, the no balancing condition produced a Macro-F1 of 0.4624 with a sarcastic class F1 of 0.0000. The confusion matrix confirms this directly - the model predicted 1,878 non-sarcastic instances correctly but classified all 305 sarcastic instances as non-sarcastic, achieving 86% accuracy while completely failing its actual purpose.

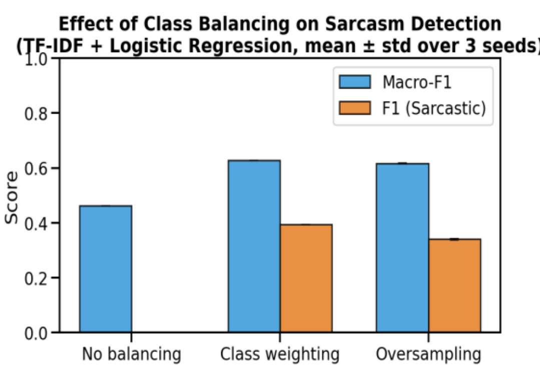

Figure 14: Effect of Class Balancing on Sarcasm Detection (TF-IDF + Logistic Regression, mean ± std over 3 seeds.

Both balancing strategies improved results substantially. Class weighting achieved a Macro-F1 of 0.6275 and a sarcastic class F1 of 0.3931 with precision of 0.32 and recall of 0.52, showing the model now captures over half of all sarcastic instances. Oversampling achieved a comparable Macro-F1 of 0.6156 but a slightly lower sarcastic F1 of 0.3397 with small variance across seeds (std of 0.002). Notably, class weighting and no balancing produced identical results across all three seeds with zero variance, which is expected since both strategies are fully deterministic given the fixed dataset splits. Only oversampling introduces randomness through its minority class duplication process. Based on these findings, class weighting is the recommended strategy for downstream experiments where the model architecture supports it, as it achieves the highest sarcastic class recall at 0.52 with no instability. For models that do not natively support class weighting, random oversampling is a suitable alternative. These findings directly inform the training choices made in Sections 2.1, 2.2, and 2.3.

Setting	Macro-F1 (mean)	Macro-F1 (std)	F1 Sarcastic (mean)	F1 Sarcastic (std)
No balancing	0.4624	0.0	0.0	0.0
Class weighting	0.6275	0.0	0.3931	0.0
Oversampling	0.6156	±0.002	0.3397	±0.003

Table 3: Balancing Strategy Summary (mean ± std over 3 seeds).

2. Experimentation with three different setups

2.1 Baseline/PTLM Gap (Sentiment/Sarcasm)

2.1.1 Experimental Setup

This section compares three categories of text classification models on both Sentiment and Sarcasm tasks of the BESSTIE corpus [1]: (1) classical TF-IDF baselines, (2) a shallow embedding baseline, FastText [18], and (3) a fine-tuned transformer. The goal is to measure the gap between the Baseline and the Pre-Trained Language Model (PTLM) on each task.
Each model is trained on the pooled BESSTIE training set (all three varieties) and tested on the combined test set of 2,183 samples. No test instances are seen during training or model selection. Three random seeds (42, 123, 456) are used per setup, and every reported metric is the mean ± standard deviation over those runs. To address the sarcasm class imbalance identified in Section 1.1, three strategies are applied (one per model family): balanced class weights for the classical baselines, minority-class oversampling for FastText (which does not support a class-weight parameter), and inverse-frequency class weights in the RoBERTa cross-entropy loss.

2.1.2 Model Configurations

The classical baselines turn the text to TF-IDF features (unigrams and bigrams, sublinear term-frequency scaling, max_features=50,000, min_df=2, max_df=0.95). These features go into two models. The first is Logistic Regression with the lbfgs solver and max_iter set to 1,000. The second is LinearSVC. The FastText supervised baseline [18] learns shallow subword embeddings. Its hyperparameters are autotuned on the validation set for 120 seconds per seed. The transformer baseline is RoBERTa-base [5] (which it has125M parameters) is fine-tuned with a linear classification head using AdamW (learning rate 2e-5, weight decay 0.01, warmup ratio 0.1, batch size 16, maximum sequence length 128 tokens) for up to 5 epochs, with early stopping on validation Macro-F1 (patience 2).

2.2 Encoder Variations and Cross-Variety Evaluation

This section presents the design, experimental setup, and evaluation of three pre-trained Transformer encoder models fine-tuned on the BESSTIE dataset [1] for binary Sentiment and Sarcasm classification across three varieties of English: Australian (en-AU), Indian (en-IN), and British (en-UK). The models evaluated are microsoft/deberta-v3-base [19], google/rembert [20], and xlm-roberta-base [12]. All experiments were conducted across three random seeds (42, 123, 456) to ensure statistical reliability, following the requirement to run at least two training runs per setup [21].

2.2.1 Models Selected

Three encoder-only Transformer architectures were selected to examine how pre-training objectives, vocabulary, and model capacity affect dialectal generalisation: **DeBERTa-v3-base (microsoft/deberta-v3-base) [19]:** DeBERTa v3 improves upon BERT and RoBERTa by introducing disentangled attention, where word content and positional embeddings are kept separate, and an enhanced mask decoder for MLM pre-training. DeBERTa-v3 additionally uses ELECTRA-style replaced token detection (RTD) on top of the disentangled attention, yielding strong single-language English representations. With ~86M parameters, it is well-suited to English-centric classification tasks. **RemBERT (google/rembert) [20]:** RemBERT is a large multilingual encoder (~559M parameters) pre-trained on 110 languages. Unlike mBERT, RemBERT decouples the input and output embedding sizes, using smaller input embeddings and substantially larger output embeddings, which improves cross-lingual transfer performance. Its multilingual pre-training makes it particularly apt for handling the code-mixed and dialectally diverse text in the BESSTIE dataset. **XLM-RoBERTa-base (xlm-roberta-base) [12]:** XLM-RoBERTa is a multilingual variant of RoBERTa trained on 100 languages using the Common Crawl corpus (~100B tokens). With ~278M parameters, it applies an expanded SentencePiece vocabulary and Masked Language Modelling (MLM) across diverse languages, making it another strong baseline for multilingual and multi-dialect sequence classification. In this work, XLM-RoBERTa was evaluated exclusively on the Sentiment task.

2.2.2 Fine-Tuning Strategy: LoRA vs. Full Fine-Tuning

To determine the optimal fine-tuning approach, DeBERTa and RemBERT were each evaluated under two strategies: (1) full fine-tuning, where all model parameters are updated, and (2) Low-Rank Adaptation (LoRA) [6], a parameter-efficient method that trains only 0.1-1% of total parameters while keeping original weights frozen (see Subsection 2.3.1 for full technical details). Following the comparison, LoRA was selected as the fine-tuning method for both DeBERTa-v3-base and RemBERT across all subsequent experiments. XLM-RoBERTa was fine-tuned using the standard full fine-tuning approach, as no LoRA comparison was conducted for this model. The LoRA configuration used rank r=16, scaling factor alpha=32, and dropout=0.1, applied via the HuggingFace PEFT library [22].

2.2.3 Experimental Setup and Hyperparameters

Table 23 in Appendix summarises the hyperparameters used across all three models. All models tokenised input texts to a maximum length of 128 tokens. Training used HuggingFace's Trainer API [22] with an early stopping patience of 3 epochs evaluated on validation macro-F1. Three experimental training modes were adopted for each model-task pair, mirroring the cross-variety evaluation protocol mandated by the coursework specification [21]: **Pooled (all-variety):** All three variety training splits are concatenated, and a single model is trained and evaluated on the test set. This assesses overall model performance regardless of dialect. **Combined (per-variety):** Each variety's training split is used independently to train a separate model, which is then tested on that same variety's test set. This represents the in-variety upper bound. **Cross-variety:** A model is trained on one variety's training split and tested on all three varieties' test sets, yielding a 3×3 transfer matrix. This directly quantifies variety-to-variety generalisability. Class imbalance in the Sarcasm task (sarcastic examples are a minority class in naturalistic data [21]) was addressed through weighted loss of the minority class in the training set for all three expeirments.

2.2.4 Evaluation

All results are reported using Macro-F1 as the primary metric, consistent with the task specification [21]. Accuracy and ROC-AUC are reported as supplementary metrics. Per-class precision and recall are included to diagnose class-imbalance effects, particularly for the minority sarcastic class. Confusion matrices for the best-performing model (RemBERT) are provided in Appendix Figure 33. Results represent averages over three random seeds unless otherwise stated.

2.2.4.1 Pooled (All-Variety) Evaluation

Table 4 reports pooled test-set performance for all models. RemBERT achieves the highest macro-F1 on both tasks: 0.9074 (±0.0043) for Sentiment and 0.7212 (±0.0063) for Sarcasm, demonstrating that its large multilingual pre-training corpus provides robust English dialectal representations. XLM-RoBERTa achieves 0.8886 (±0.0026) on Sentiment, competitive with DeBERTa (0.8425) while exhibiting the lowest variance across seeds, suggesting greater training stability. DeBERTa obtains the lowest Sentiment F1 among the three models, though its narrow confidence interval (±0.0040) confirms consistent behaviour.

Model	Task	Macro-P	Macro-R	Macro-F1	Accuracy	ROC-AUC	Std (F1)
DeBERTa-v3-base	Sentiment	0.8425	0.8425	**0.8425**	0.8426	0.9151	±0.0040
	Sarcasm	0.6313	0.7493	**0.6290**	0.7230	0.8136	±0.0137
RemBERT	Sentiment	0.9074	0.9075	**0.9074**	0.9075	0.9646	±0.0043
	Sarcasm	0.7009	0.7554	**0.7212**	0.8468	0.8770	±0.0063
XLM-RoBERTa-base	Sentiment	0.8889	0.8884	**0.8886**	0.8887	0.9552	±0.0026

Table 4: Pooled Test Set Results: Macro-Precision, Macro-Recall, Macro-F1 (Mean ± Std, 3 Seeds) Macro-P and Macro-R are macro-averaged precision and recall. For Sentiment, P≈R≈F1 due to balanced classes; for Sarcasm, R > P indicating bias toward predicting the majority class.

Sarcasm F1 scores are substantially lower than Sentiment F1 scores across both models. DeBERTa reaches only 0.6290 on the pooled sarcasm test set, and RemBERT achieves 0.7212. This gap is expected: sarcasm is a pragmatic, context-heavy phenomenon whose detection requires cultural and discourse-level knowledge that standard MLM pre-training may not adequately capture [23]. The high ROC-AUC scores (>0.80 for both models) indicate that models can rank sarcastic instances above non-sarcastic ones probabilistically, but struggle to convert this ranking into reliable binary predictions at the per-class level, particularly for the minority sarcastic class, as discussed in Subection 3.3. The gap between Macro-Precision and Macro-Recall for the Sarcasm task (Table 6) further confirms this: both models exhibit Recall exceeding Precision by 8-12 points, indicating a tendency to over-predict the majority non-sarcastic class.

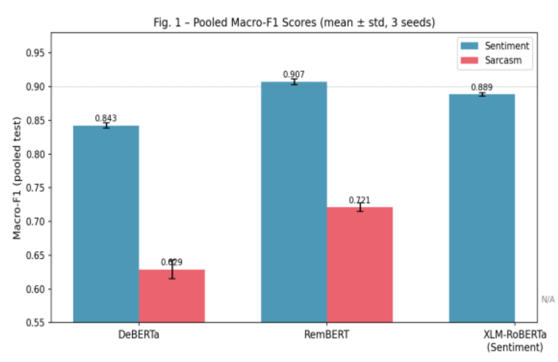

Figure 15: Pooled Macro-F1 Scores (mean ± std over 3 seeds). Error bars show standard deviation. XLM-RoBERTa evaluated on Sentiment only.

2.2.4.2 Per-Variety Combined Results

Table 5 and Table 6 present per-class precision and recall, along with macro-F1 and accuracy, for each model trained and evaluated on individual variety splits.

2.2.4.2.1 Sentiment

Model	Variety	Neg-P	Neg-R	Neg-F1	Pos-P	Pos-R	Pos-F1	Macro-P	Macro-R	Macro-F1
DeBERTa	en-AU	0.817	0.898	**0.855**	0.876	0.781	**0.826**	0.844	0.840	**0.841**
	en-IN	0.821	0.788	**0.804**	0.774	0.808	**0.791**	0.797	0.798	**0.797**
	en-UK	0.914	0.867	**0.890**	0.880	0.923	**0.901**	0.897	0.895	**0.895**
RemBERT	en-AU	0.896	0.924	**0.910**	0.915	0.883	**0.899**	0.905	0.904	**0.904**
	en-IN	0.884	0.848	**0.866**	0.838	0.876	**0.856**	0.861	0.862	**0.861**
	en-UK	0.968	0.958	**0.963**	0.961	0.970	**0.965**	0.965	0.964	**0.964**
XLM-RoBERTa	en-AU	0.864	0.925	**0.893**	0.912	0.842	**0.875**	0.888	0.884	**0.884**
	en-IN	0.860	0.851	**0.855**	0.836	0.845	**0.841**	0.848	0.848	**0.848**
	en-UK	0.941	0.933	**0.937**	0.938	0.944	**0.941**	0.939	0.939	**0.939**

Table 5: Per-Variety Sentiment: Per-Class and Macro Precision, Recall, F1 (Avg. 3 Seeds) Neg = Negative class (label 0); Pos = Positive class (label 1). Macro-P and Macro-R are the unweighted averages across both classes.

A consistent variety ordering is observed across all three models: en-UK (British English) yields the highest macro-F1, followed by en-AU (Australian English), with en-IN (Indian English) achieving the lowest scores. This mirrors the linguistic hierarchy in the BESSTIE dataset: en-UK and en-AU are both Inner-Circle varieties with comparable syntactic structures, while en-IN is an Outer-Circle variety characterised by code-mixed Hindi-English, dialectal grammar, and Indian-specific slang [21]. RemBERT achieves an exceptional 0.964 F1 on en-UK and 0.904 on en-AU, largely outperforming DeBERTa and XLM-RoBERTa on all varieties. This advantage is attributed to RemBERT's multilingual pre-training, which develops broader morphological representations capable of handling dialectal variation [20].

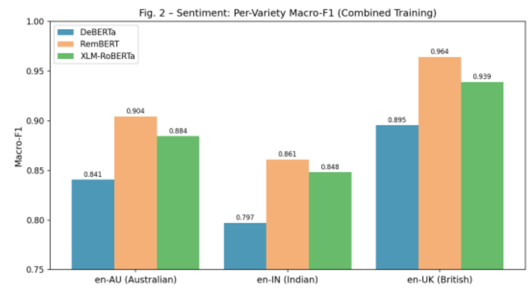

Figure 16: Sentiment Macro-F1 by variety and model (combined/per-variety training). en-UK consistently yields highest F1 across models.

2.2.4.2.2 Sarcasm

Model	Variety	Neg-P	Neg-R	Neg-F1	Pos-P	Pos-R	Pos-F1	Macro-P	Macro-R	Macro-F1
DeBERTa	en-AU	0.878	0.684	**0.768**	0.506	0.769	**0.609**	0.692	0.727	**0.688**
	en-IN	0.968	0.704	**0.814**	0.145	0.679	**0.239**	0.556	0.691	**0.527**
	en-UK	0.996	0.745	**0.852**	0.236	0.962	**0.379**	0.616	0.854	**0.616**
RemBERT	en-AU	0.856	0.866	**0.861**	0.669	0.648	**0.658**	0.762	0.757	**0.759**
	en-IN	0.953	0.913	**0.932**	0.247	0.387	**0.301**	0.600	0.650	**0.617**
	en-UK	0.982	0.858	**0.916**	0.322	0.811	**0.460**	0.652	0.835	**0.688**

Table 6: Per-Variety Sarcasm: Per-Class and Macro Precision, Recall, F1 (Avg. 3 Seeds) For sarcasm, Neg = Non-Sarcastic (majority class); Pos = Sarcastic (minority class). High Neg-P with very low Pos-P in en-IN and en-UK confirms majority-class bias.

The sarcasm task reveals a pronounced class-imbalance problem. For both models on en-IN and en-UK, the positive class (Sarcastic) achieves very low precision, DeBERTa achieves only 0.145 and RemBERT only 0.247 on en-IN, despite moderate to high recall. This indicates that the models bias toward predicting the majority class (Non-Sarcastic) and only intermittently predict sarcasm. The high Negative-class recall (0.70–0.97) paired with low Positive-class precision (0.15–0.51) is a textbook manifestation of class-imbalance bias [23]. Even with oversampling applied during training, the inherent scarcity of sarcastic instances in naturalistic data makes generalisation difficult. en-AU consistently yields the highest sarcasm F1 for both models (DeBERTa: 0.688, RemBERT: 0.759). Australian English sarcastic expressions in the BESSTIE dataset appear more learnable, possibly because Australian sarcasm often relies on overt lexical cues ("Good onya, mate.") that are more pattern-recognisable than the subtle contextual sarcasm more common in en-IN.

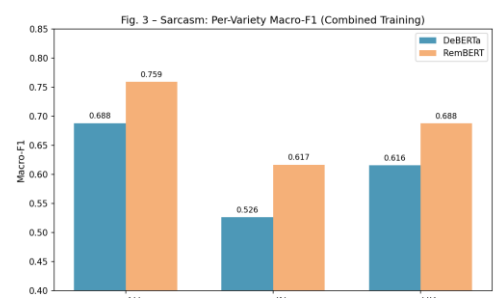

Figure 17: Sarcasm Macro-F1 by variety (DeBERTa vs RemBERT). en-IN yields the lowest F1 for both models, exposing the variety gap.

2.3 LLM Variations with LoRA Adapters (Sarcasm)

The primary objective of this section is to evaluate whether lightweight LoRA (Low-Rank Adaptation) [6] adapters, trained on top of a frozen open-weight large language model (LLM), can achieve robust sarcasm detection across three dialectally distinct English varieties: British (en-UK), Australian (en-AU) and Indian (en-IN), drawn from the BESSTIE benchmark [1]. Four architectures are systematically compared: an English-centric decoder LLM, TinyLlama-1.1B [7], adapted with LoRA; a multilingual decoder LLM, Qwen2.5-1.5B [8], adapted with LoRA; and a multilingual encoder, XLM-RoBERTa-base [12], in two variants, full fine-tuning and LoRA (Table 24 in Appendix). The dual XLM-R inclusion is deliberate: the full fine-tuning condition provides the conventionally fine-tuned PTLM baseline against which both decoder LoRA adapters and LoRA on the same encoder can be contrasted, while holding the backbone constant between the two XLM-R conditions yields a controlled ablation that isolates the effect of parameter-efficient adaptation from the effect of backbone family. Each model is trained over three independent runs per variety to ensure statistical reliability. Furthermore, due to the absence of sarcasm labels in Google subset, sarcasm classification is conducted only on reddit subset.

2.3.1 Low-Rank Adaptation (LoRA)

Full fine-tuning of LLMs is prohibitively expensive for most practitioners [4]. LoRA [6] addresses this by constraining weight updates to a low-rank subspace: for a frozen weight matrix $W \in \mathbb{R}^{\wedge}(dxk)$, the update is factored as $\Delta W = BA$, where $B \in \mathbb{R}^{\wedge}(dxr)$ and $A \in \mathbb{R}^{\wedge}(rxk)$ with rank $r \ll min(d, k)$. Only A and B are trained, and A is initialised to zero so that ΔW=0 at training onset. The scaling factor α controls the effective magnitude of the adapter's contribution: the adapted weight is $W + (\alpha/r) \cdot BA$. The low-rank assumption is theoretically motivated by the near-low-rank structure of gradient updates in over-parameterised neural networks [13]. In this implementation, LoRA is applied to the query and value projection matrices of every self-attention layer [14], **q_proj/v_proj for the decoder LLMs and query/value for XLM-RoBERTa**. This q/v-only targeting follows the original LoRA paper [6], which empirically demonstrates that attention projections query and value together **capture most of the adaptable capacity without the memory overhead of all-linear targeting**; this reduces trainable parameters to ~0.1% of the base model versus ~0.5% for all-linear QLoRA configurations, fitting the GPU budget while retaining the expressiveness needed for small-data classification. Setting α=16 with r=8 gives a scaling factor of 2, which stabilises training across different rank choices [6]. A dropout of 0.05 provides light regularisation [9]. The resulting on-disk adapter footprint is 59-92 MB depending on the backbone which is tiny compared to the ~2-3 GB base model loaded once into GPU memory and shared across all variety adapters, enabling the hot-swap deployment pattern.

2.3.2 Variety Gap and Architectural Motivation

The variety gap is quantified as the difference between in-distribution Macro-F1 (adapter trained and tested on the same variety) and the mean cross-variety Macro-F1. A large gap indicates variety-specific features that do not transfer; a small gap indicates transferable pragmatic abstractions. The surface realisations of sarcasm differ substantially across English varieties (see Introduction and Section 1.2), motivating variety-specific adaptation [2]. Sarcasm detection is a sentence-level task that benefits from full bidirectional context, giving encoder models like XLM-RoBERTa [12] (see Section 2.2.1) a structural advantage over autoregressive decoders such as TinyLlama [7] and Qwen2.5 [8], which attend only to preceding tokens [14]. However, as introduced in Section 2.1, LoRA enables decoders to leverage their generation-oriented representations for classification with only a few million trainable parameters, making the comparison empirically interesting rather than theoretically pre-determined.

2.3.3 BESSTIE Dataset and Class Imbalance

As established in Section 1.1, in BESSTIE dataset [1], sarcasm is severely under-represented across all varieties, particularly in Google Places reviews. Unlike the class-weighting approach evaluated in Section 1.3, the LLM experiments adopt a two-step strategy applied to the training split only: (i) a cap of 800 samples per variety with all minority-class instances retained and the majority class down-sampled, and (ii) oversampling of the minority class with replacement to a 1:1 ratio. Validation and test splits are untouched. This was sufficient to prevent majority-class collapse across all four architectures and all varieties; Macro-F1 remains the primary metric to penalise degenerate majority-class predictions [9], [10]. See Figure 18 for the post-oversampling distribution.

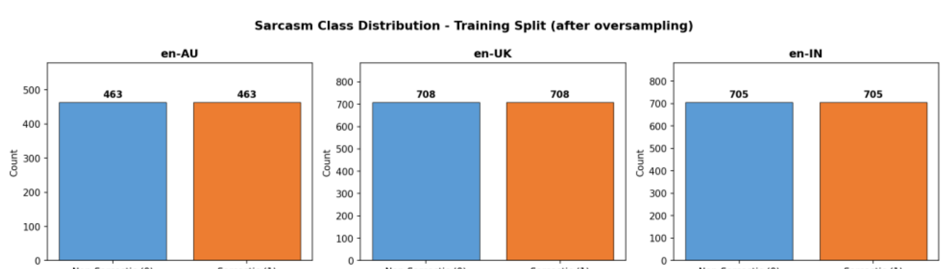

Figure 18: Sarcasm class distribution per variety in the training split after oversampling. Sarcastic and Non-Sarcastic counts are equalised so that Macro-F1 rather than accuracy drives model comparison.

2.3.4 Experimental Methodology

2.3.4.1 Architectures

TinyLlama-1.1B [7] and **Qwen2.5-1.5B** [8] are adapted with identical LoRA configurations to enable a controlled comparison of English-dominant versus multilingual decoder pre-training. **XLM-RoBERTa-base** [12] is included in two variants, **full fine-tuning and LoRA**, which is methodologically deliberate. First, the full-FT condition satisfies Section 2.1 of the coursework brief by providing a conventionally fine-tuned pre-trained encoder as the PTLM baseline against which the decoder LLM adapters can be contrasted. Without it, any comparison between decoders and an encoder would conflate the effect of backbone family with the effect of adaptation strategy. Second, holding the backbone constant (XLM-R-base) and varying only the adaptation strategy yields a controlled ablation: any difference between the two XLM-R conditions is attributable solely to LoRA, since pre-trained weights, tokenizer, data, seeds, epochs, learning rate and batch size are identical. This design answers two questions simultaneously: "does a multilingual encoder beat LLM decoders on varieties of English?" and "does LoRA match full fine-tuning in the small-data regime?". All four backbones use byte-pair encoding tokenisers, ensuring consistent sub-word segmentation across the three English varieties.

2.3.4.2 LoRA Configuration and Training Protocol

LoRA hyperparameters are held constant across all three LoRA configurations: rank r=8, α=16 (scaling = 2), dropout = 0.05, target modules q_proj/v_proj for decoders and query/value for XLM-RoBERTa, learning rate 2×10^{-4}, batch size 8(train) / 16(eval), max_length=128 tokens (covering>95% of BESSTIE sequence lengths), and BF16 (Ampere+) or FP16 precision auto-detected at runtime. Training runs for up to 3 epochs with early stopping (patience 2). XLM-RoBERTa full fine-tuning uses an identical schedule but updates all 278M parameters. Three runs per variety (seeds 43/44/45) are conducted; the best checkpoint by validation Macro-F1 is retained. A pooled combined adapter is also trained on the union of the three balanced training splits. Evaluation is performed on all four test sets (en-AU, en-UK, en-IN, combined) forming a 4x4 cross-variety matrix per architecture, with Macro-F1 as the primary metric plus per-class F1, precision, recall, confusion matrices and inference latency (ms/sample, GPU-synchronised). Zero-shot baselines use the frozen base model with a randomly initialised classification head.

2.3.4.3 LoRA Experiment: Perplexity and Pre-Training Calibration

Perplexity is the exponentiated mean negative log-likelihood that a language model assigns to a sequence: $PPL = exp(-(1/N)\ \Sigma_{\text{i}}\ log\ p(x_i|x_{<i}))$. With no fine-tuning, perplexity on a target variety's text directly **indicates how well that variety is covered by the backbone's pre-training distribution**, without any classification head involved. Comparing TinyLlama (English-only) and Qwen2.5 (multilingual, 29 languages) on BESSTIE text isolates the effect of pre-training mixture from any downstream task signal.

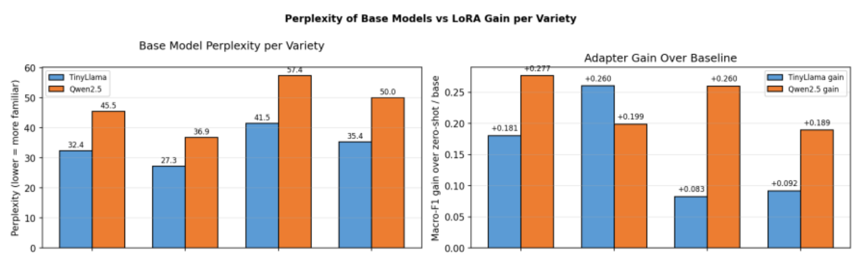

Figure 19: Base-model perplexity vs. LoRA gain over zero-shot for TinyLlama-1.1B and Qwen2.5-1.5B. Lower perplexity on the target variety correlates with a larger downstream gain, especially on en-AU.

Three results stand out: (i) TinyLlama exhibits 38–41% lower perplexity than Qwen2.5 on every variety despite being 27% smaller, confirming that multilingual breadth comes at the cost of per-language depth. (ii) Both models rank en-UK < en-AU < en-IN consistently, confirming en-IN is the hardest variety regardless of backbone. (iii) The en-IN perplexity gap is the largest for both LLMs, validating that en-IN is genuinely out-of-distribution, not just harder to classify, but harder to model, supporting Kachru's outer-circle classification [2].

This perplexity gap explains the downstream pattern: TinyLlama outperforms Qwen2.5 on three of four in-distribution cells (en-AU +0.0126, en-UK +0.0735, combined +0.0203 Macro-F1), with Qwen2.5 winning only on en-IN (+0.0161), even though the multilingual model is 36% larger and explicitly covers Hindi. This finding holds at the 1–1.5B scale but should not be generalised unconditionally: at much larger scales, Srirag et al. [1] report that multilingual models substantially outperform monolingual ones on the same dataset, suggesting multilingual breadth becomes advantageous once capacity is sufficient to preserve per-language depth. Results for all four architectures, including per-class metrics, the LoRA vs full fine-tuning ablation, and cross-variety transferability analysis, are presented in Sub-section 3.3, with supporting confusion matrices in the Appendix.

Model/Variety	Perplexity			
	en-AU	en-UK	en-IN	combined
TinyLlama-1.1B before LoRA	32.43	**27.31**	**41.52**	35.38
Qwen2.5-1.5B before LoRA	45.49	36.87	**57.37**	50.02

Table 7: Base-model perplexity (frozen, no adapter) per variety. Lower is better. The monolingual TinyLlama trained on UK English shows the lowest Perplexity value for en-UK as expected.

3 Evaluation

3.1 Baseline/PTLM Gap Results

Table 8 reports Macro-F1, Precision, and Recall (mean ± standard deviation over three seeds) for all 4 model setups on both tasks. shows the main comparison across the three model families.

Task	Model	Macro-F1	Precision	Recall
Sentiment	TF-IDF + Logistic Regression	0.8287 ± 0.0000	0.8300 ± 0.0000	0.8285 ± 0.0000
Sentiment	TF-IDF + Linear SVM	0.8407 ± 0.0000	0.8418 ± 0.0000	0.8404 ± 0.0000
Sentiment	FastText (supervised)	0.7991 ± 0.0227	0.8008 ± 0.0223	0.8000 ± 0.0226
Sentiment	RoBERTa-base (fine-tuned)	0.8915 ± 0.0054	0.8917 ± 0.0054	0.8916 ± 0.0053
Sarcasm	TF-IDF + Logistic Regression	0.6275 ± 0.0000	0.6143 ± 0.0000	0.6688 ± 0.0000
Sarcasm	TF-IDF + Linear SVM	0.5932 ± 0.0000	0.6103 ± 0.0000	0.5838 ± 0.0000
Sarcasm	FastText (supervised)	0.5950 ± 0.0189	0.6087 ± 0.0029	0.6051 ± 0.0470
Sarcasm	RoBERTa-base (fine-tuned)	0.7004 ± 0.0096	0.6805 ± 0.0093	0.7365 ± 0.0124

Table 8: Macro-F1, Precision, and Recall for the three model families on the Sentiment and Sarcasm tasks of BESSTIE, pooled across all three varieties.The values are the mean and standard deviation over three random seeds (42, 123, 456). The classical models show no seed variance because their solvers give the same result every time once the feature matrix is fixed. FastText and RoBERTa show some variance, which comes from the auto tuning step and the random weight setup.

Pre-training gives a clear gap on both tasks. On Sentiment, the best classical baseline (TF-IDF with Linear SVM) achieves a Macro-F1 of 0.8407 and RoBERTa-base gets 0.8915. This is an absolute gap of +0.0508 and a relative gap of +6.0%. On Sarcasm, the best classical baseline (TF-IDF with Logistic Regression) achieves 0.6275 and RoBERTa achieves 0.7004. That is an absolute gap of +0.0729 and a relative gap of +11.6%. So the relative gap is about twice as large on the pragmatic Sarcasm task as on the lexical Sentiment task. This fits the idea that the main thing pre-training's adds is contextual reasoning not just spotting the polarity of single words [3].

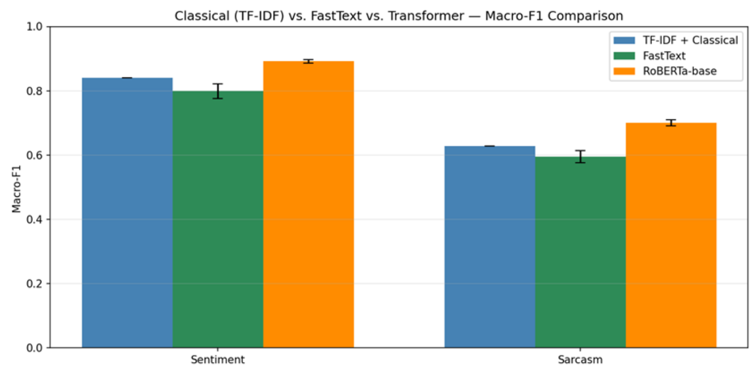

Figure 20: Macro-F1 comparison of the classical TF-IDF baselines, the FastText shallow embedding baseline, and the fine tuned RoBERTa base transformer on the Sentiment and Sarcasm tasks. The error bars show one standard deviation above and below the mean over three seeds.

In addition, FastText does worse than the best TF-IDF baseline on both tasks. It scores 0.7991 against 0.8407 on Sentiment and 0.5950 against 0.6275 on Sarcasm. This result is interesting because FastText learns subword embeddings, while TF-IDF is just sparse word counts. But FastText loses the explicit bigram negation signal that TF-IDF keeps. Its way of handling class imbalance through oversampling, which is less effective than the class-weight reweighting used by the classical models. Learned embeddings are not always better than engineered features for this task. What matters is the interaction between the feature space, the loss function, and the class-imbalance treatment.

3.1.1 Per-Class and Per-Variety Analysis

A single Macro F1 score can hide a model that just gets better at the majority class while still failing to spot sarcasm [9]. In Figure 21 the sarcastic class in the test split has only 305 of 2,183 samples, which is 14.0%. and thus, it is the most discriminating cell of the evaluation. On this class, the best classical Logistic Regression model gets a precision of 0.315, a recall of 0.521, and an F1 0.393. So it spots 52.1% of the truly sarcastic samples. RoBERTa base improves these to a precision of 0.418, a recall of 0.639, and an F1 0.506. That is an absolute recall gain of +0.118 and a relative recall gain of +22.6%. The key point is that neither model collapses onto the majority class. Both detect sarcasm, which is the minimum bar for reporting sarcasm classification as a real capability rather than a majority-class artefact.

If we look at the Sarcasm confusion matrix in Figure 21, RoBERTa beats the best classical baseline in every cell. True Positives go up from 159 to 195. False Positives drop from 345 to 271. False Negatives drop from 146 to 110. So the recall gain is a real win across every type of error, not a trade off between precision and recall. All three model groups share the same kind of error pattern on Sarcasm. False Positives outnumber False Negatives by about two to one. This is the typical sign of inverse frequency class weighting, which pushes the model to over predict the minority class to make up for the 14% prior. None of the models fall into the other failure mode where sarcasm is just ignored. The per class errors are shown as confusion matrices in Figure 21.

Although trained on pooled data, the model may not perform uniformly across varieties. Figure 22 reports per-variety test-set Macro-F1 for en-AU, en-IN, and en-UK to verify this. On Sentiment, the gap is positive and fairly steady across all three varieties. It is 0.062 on en-AU, 0.040 on en-IN, and 0.044 on en-UK. On Sarcasm, the picture is very different. The gap is 0.096 on en-AU, which is the biggest gain in this section. It is 0.071 on en-UK, which is moderate. And it is 0.004 below zero on en-IN. So the en-IN cell is a small regression where the classical Logistic Regression model matches RoBERTa on Indian English sarcasm within seed variation.

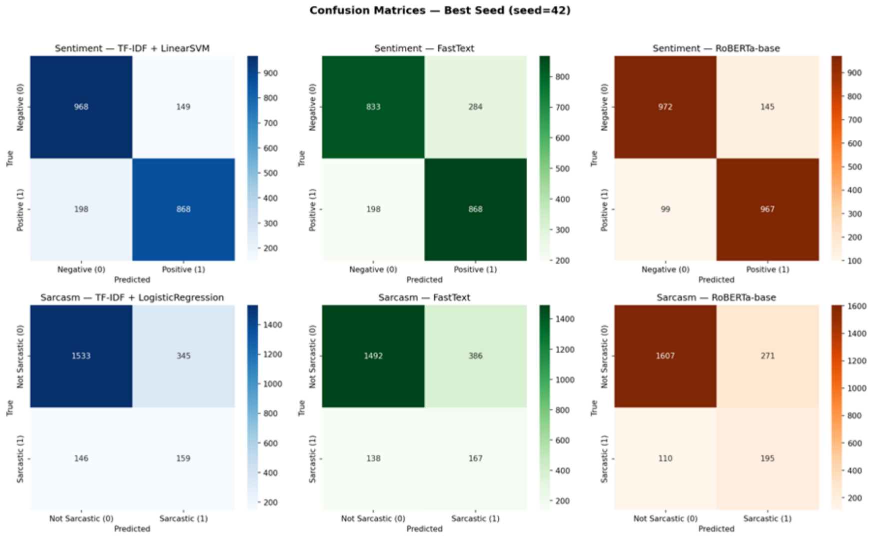

Figure 21: Confusion matrices for the best classical TF-IDF classifier (left column), FastText (centre column), and RoBERTa-base (right column) on the Sentiment task (top row) and the Sarcasm task (bottom row). Each cell uses a sequential blue/green/orange colour.

One way to read this is that en-IN, as an Outer Circle variety [2] with mixed Hindi and English vocabulary, leans more on clear lexical cues. These include code mixed interjections, emphatic particles taken from Hindi, and sarcastic templates that are specific to the variety. A well tuned bag of bigrams model can pick up these cues just as well as the subword tokeniser of RoBERTa. The two Inner Circle varieties, on the other hand, rely more on implicit pragmatic contrasts. For these, the bidirectional attention of RoBERTa over the full sentence gives the expected gain from context.

3.1.2 Discussion

The results show a clear and steady PTLM gap on both BESSTIE tasks, but two things are worth pointing out. First, the gap is larger in relative terms on the pragmatic Sarcasm task than on the lexical Sentiment task. Sentiment leans heavily on clear polar words like "excellent" or "terrible", which a well built bag of bigrams model already picks up. So the leftover Macro F1 gap of about 0.05 only reflects the borderline cases where context attention shifts the decision. Sarcasm, on the other hand, needs the model to track the tension between the surface meaning of the words and the real intent [3]. This is exactly what bidirectional attention [14] over the full sentence gives you, which is why the gap is about twice as wide.

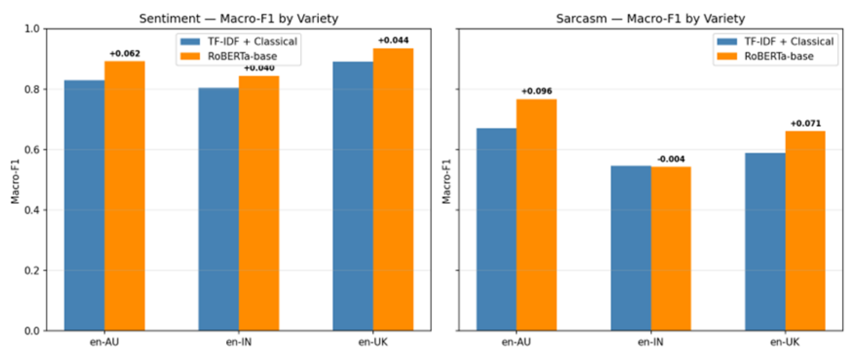

Figure 22: Per variety Macro F1 for the best classical TF-IDF baseline and the fine tuned RoBERTa base on the Sentiment and Sarcasm tasks (seed=42). The RoBERTa advantage is positive and steady on Sentiment, but on Sarcasm there is an Outer Circle reversal on en-IN, where the classical model is tied with the transformer.

A few examples from the Sarcasm test set show the nature of this gap. The sentence "I for one welcome our new goose overlord" (en-AU, true label Sarcastic) is picked up by RoBERTa but missed by the TF-IDF baseline. No single word here is openly negative. The sarcasm comes from the ironic "welcome our new overlord" template, a pragmatic pattern. To spot it, you have to see the contrast between the fake positive surface and the speaker's real disapproval. TF-IDF treats each n-gram on its own, so it only sees positive and neutral words and predicts Non Sarcastic. The other direction also happens. The sentence "Both parties are fucking useless let's be honest, dismantle parliament" (en-UK, true label Non Sarcastic) is picked up by TF-IDF but flagged as sarcastic by RoBERTa. Here the strong language is real frustration, not irony. The direct lexical match used by TF-IDF, where openly negative words mean negative intent, happens to be the right call. RoBERTa seems to read the dramatic tone as a sign of sarcasm, which fits the False Positive bias seen in the confusion matrices (Figure 21).

The second point is that the Sarcasm gap is not the same across varieties. The RoBERTa advantage is mainly on the Inner Circle varieties (en-AU and en-UK) and almost goes away on the Outer Circle en-IN variety. Two reasons fit this pattern, and they can both be true at once. The first is coverage. The pre training data of RoBERTa has less Indian English text than British or Australian text, so its context representations for en-IN code mixing may just be weaker than for the Inner Circle varieties. The second is that en-IN sarcasm in BESSTIE leans more on clear lexical markers, which sparse feature models can pick up just as well. Telling these two reasons apart is beyond the scope of this section, but Section 2.2 (cross variety transfer) and Section 2.3 (per variety LoRA tuning) give more evidence on the question. The confusion matrix analysis also shows that the leftover Sarcasm errors of

RoBERTa stay dominated by False Positives (271 vs 110 False Negatives), indicating that the inverse-frequency loss weighting still pushes the model to over-predict sarcasm even after fine-tuning.

3.2 Cross-variety Evaluation Results

Cross-variety evaluation trains each model on one variety and evaluates on all three, producing a 3×3 transfer matrix. Diagonal entries reflect in-variety performance; off-diagonal entries capture variety-to-variety transfer. The key question is whether models trained on Inner-Circle varieties (en-UK, en-AU) transfer better to each other than to the Outer-Circle variety en-IN.

3.2.1 Sentiment Transfer

RemBERT exhibits remarkably strong cross-variety sentiment transfer: even when trained exclusively on en-IN, it achieves 0.8991 on en-AU and 0.9562 on en-UK. This near in-variety performance (0.8608 in-variety for en-IN) demonstrates that the variety gap for Sentiment is largely bridged by RemBERT's multilingual representations. XLM-RoBERTa also transfers well, with all off-diagonal values above 0.80, but DeBERTa shows more pronounced degradation: training on en-IN and testing on en-AU drops to 0.6285, a 7-point gap below the in-variety 0.6199 score, indicating that DeBERTa's English-centric pre-training provides less dialectal robustness.

A consistent pattern emerges: testing on en-UK consistently yields higher F1 than testing on en-AU or en-IN, regardless of the training variety. This is consistent with en-UK being closer to the formal written English present in most pre-training corpora. Testing on en-IN invariably yields the lowest cross-variety performance, confirming the greater linguistic distance of Indian English from the other two varieties [21].

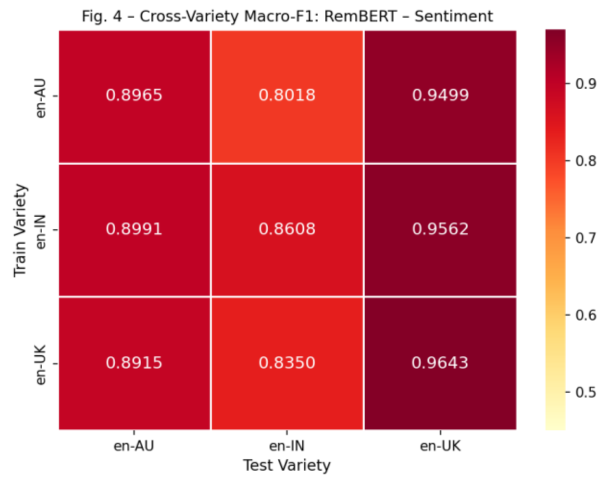

Figure 23: Cross-variety Macro-F1 matrix for RemBERT Sentiment. Green diagonal = in-variety. Row = Train variety, Column = Test variety.

Train Variety	Test Variety	RemBERT			DeBERTa			XLM-RoBERTa		
		Macro-P	Macro-R	Macro-F1	Macro-P	Macro-R	Macro-F1	Macro-P	Macro-R	Macro-F1
en-AU	**en-AU**	**0.8992**	**0.8959**	**0.8965**	**0.7183**	**0.7169**	**0.7171**	**0.8882**	**0.8848**	**0.8855**
	en-IN	0.8064	0.8042	0.8018	0.7038	0.7020	0.6983	0.8079	0.8085	0.8071
	en-UK	0.9502	0.9498	0.9499	0.8032	0.7885	0.7882	0.9261	0.9243	0.9246
en-IN	en-AU	0.9016	0.8982	0.8991	0.6379	0.6339	0.6285	0.8798	0.8729	0.8740
	en-IN	**0.8608**	**0.8610**	**0.8608**	**0.6342**	**0.6286**	**0.6199**	**0.8450**	**0.8443**	**0.8443**
	en-UK	0.9561	0.9564	0.9562	0.6954	0.6718	0.6648	0.9330	0.9321	0.9323
en-UK	en-AU	0.8995	0.8900	0.8915	0.7527	0.7395	0.7389	0.8700	0.8627	0.8638
	en-IN	0.8351	0.8352	0.8350	0.7299	0.7235	0.7235	0.8146	0.8149	0.8145
	en-UK	**0.9643**	**0.9643**	**0.9643**	**0.8434**	**0.8399**	**0.8402**	**0.9291**	**0.9275**	**0.9279**

Table 9: RemBERT, DeBERTa and XML-RoBERTa Cross Variety Sentiment Transfer (Macro-P, Macro-R, Macro-F1). Bold values indicate in-variety results where train and test variety are identical.

3.2.2 Sarcasm Transfer

Cross-variety sarcasm transfer is substantially weaker than sentiment transfer for both models. DeBERTa trained on en-AU achieves only 0.4742 on en-IN, a drop of 12 points from the in-variety score of 0.5737 for en-IN. RemBERT trained on en-AU achieves 0.4891 on en-IN, confirming that Australian sarcasm patterns do not generalise to Indian English contexts. This is linguistically expected: sarcasm is deeply pragmatic and culturally embedded, relying on shared referential frames (e.g., Australian slang "Good onya" being sarcastic; Indian code-mixed sarcasm invoking cultural shorthand) that models trained on one variety cannot easily acquire [23].

Notably, training on en-UK yields slightly better transfer to en-AU (0.6398 for RemBERT) than training on en-IN (0.5156), confirming the Inner-Circle proximity hypothesis: British and Australian varieties share sufficient surface-level sarcasm markers that en-UK training partially benefits en-AU

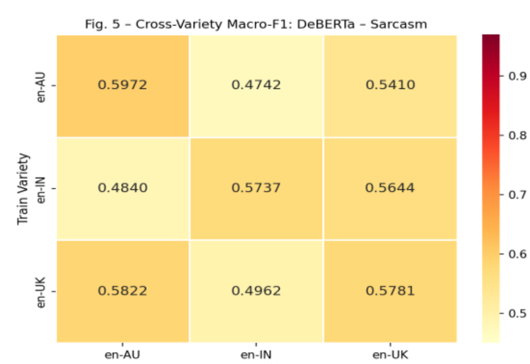

Figure 24: Cross-variety Macro-F1 matrix for DeBERTa Sarcasm. Off-diagonal drops are substantial, confirming that sarcasm is variety-sensitive.

evaluation. However, both Inner-Circle trained models transfer poorly to en-IN sarcasm (0.49–0.56 range), exposing a clear Outer-Circle/Inner-Circle sarcasm gap.

Train Variety	Test Variety	DeBERTa			RemBERT		
		Macro-P	Macro-R	Macro-F1	Macro-P	Macro-R	Macro-F1
en-AU	**en-AU**	**0.5983**	**0.6080**	**0.5972**	**0.7440**	**0.7759**	**0.7528**
	en-IN	0.5339	0.6252	0.4742	0.5415	0.6524	0.4891
	en-UK	0.5502	0.6411	0.5410	0.6021	0.8155	0.5941
en-IN	en-AU	0.5591	0.5187	0.4840	0.6069	0.5435	0.5156
	en-IN	**0.5662**	**0.6313**	**0.5737**	**0.5852**	**0.6932**	**0.5986**
	en-UK	0.6062	0.5583	0.5644	0.6232	0.5989	0.6022
en-UK	en-AU	0.5938	0.5798	0.5822	0.6567	0.6325	0.6398
	en-IN	0.5333	0.6133	0.4962	0.5760	0.7487	0.5615
	en-UK	**0.5682**	**0.6336**	**0.5781**	**0.6628**	**0.8010**	**0.7001**

Table 10: DeBERTa and RemBERT Cross Variety Sarcasm Transfer (Macro-P, Macro-R, Macro-F1). Bold values indicate in-variety results where train and test variety are identical.

3.2.3 Discussion

Across all experiments, RemBERT consistently outperforms DeBERTa and XLM-RoBERTa for both tasks and all three variety settings. This supports the hypothesis that multilingual pre-training on typologically diverse languages provides representations that better accommodate dialectal variation within English, even when no explicit dialect-specific training is performed [20]. RemBERT's parameter count (~559M) is substantially larger than DeBERTa (~86M) and XLM-RoBERTa (~278M), which also contributes to its superior performance, though model size alone cannot explain the gains in cross-variety transfer.

The Sentiment task is substantially more tractable than Sarcasm for all models. Sentiment polarity maps onto relatively stable lexical-semantic features (positive vs. negative vocabulary) that generalise across varieties, whereas sarcasm detection requires pragmatic reasoning, discourse context, and cultural knowledge that vary significantly across English varieties [23]. This fundamental difference in task complexity manifests in the 15–20 point macro-F1 gap between Sentiment and Sarcasm performance across all models.

Indian English (en-IN) consistently represents the hardest evaluation variety for all models and both tasks. This can be attributed to: (i) code-mixed Hindi-English tokens that are underrepresented even in multilingual pre-training corpora, (ii) dialectal grammar and syntactic patterns that deviate significantly from standard written English, and (iii) culturally specific sarcasm markers (e.g., "Coz we all have free internet") that require world knowledge absent from text-based pre-training. Addressing this variety gap likely requires dialect-specific pre-training, data augmentation, or few-shot adaption strategies beyond standard fine-tuning [24].

Error analysis of the top-5 false positives and false negatives by model confidence (see from Table 25 to Table 28 in Appendix) reveals consistent, interpretable failure modes for both DeBERTa and RemBERT. For the Sarcasm task, DeBERTa is misled by surface-level contrast structures (mixed-sentiment reviews, concessive phrases) and non-standard linguistic input (code-mixed text, formal sign-offs), while missing sarcasm expressed through absurdist dialogue, stage-direction formatting, or culturally embedded political satire that requires world knowledge unavailable from text-based pre-training alone [24]. RemBERT, despite higher overall F1, exhibits different failure modes: its high sensitivity to learned typographic sarcasm markers (e.g., mock-capitalisation, scare-quotes) causes false positives on literal text that shares stylistic features with sarcasm, while it misses cross-lingual irony (e.g., Hindi proverbs) and deadpan narratives where absurd details are embedded in otherwise plausible prose. For the Sentiment task, both models are systematically confused by mixed-register reviews, particularly Indian-English text (en-IN) where dialectal phrasing, code-mixing, and understated positivity diverge from the predominantly inner-circle English patterns in pre-training corpora [21].

3.3 LoRA Experiment

3.3.1 LoRA Experiment: Per-Model Results

This section reports the three per-model result comparisons of training stability, in-distribution test performance (including the pooled combined condition) and the adaptation-gain analysis over each model's own zero-shot baseline, for all four architectures in turn.

3.3.1.1 LoRA Experiment: Training Stability

Validation Macro-F1 across three seeds per variety is reported per architecture below. The spread (max - min) is the stability proxy; all checkpoints selected are the per-variety maximum.

TinyLlama-1.1B (LoRA): en-AU and en-UK exhibit small spreads (0.0144 and 0.0203), indicating reliable convergence; en-IN shows a larger spread of 0.0522, reflecting greater optimisation sensitivity on the imbalanced outer-circle variety [9]. All runs exceed chance, confirming no run collapsed to the majority-class prediction.

Qwen2.5-1.5B (LoRA): en-UK shows the largest spread (0.0905), Run 2 substantially outperforms Runs 1 and 3, consistent with a noisier training signal under the lower en-UK Reddit sarcasm prevalence. en-IN shows zero spread, all three runs converging to exactly 0.6961; this is analytically unusual and likely reflects early stopping triggering at the same epoch each time under class-imbalance dynamics [9].

XLM-RoBERTa-base (Full FT): en-AU shows the largest spread (0.1167), with Run 1 substantially under-performing, high initialisation sensitivity when updating all 278 M parameters on only 800 samples, a known overfitting failure mode that motivates the LoRA ablation [6], [4]. en-IN is the most stable (0.0265).

XLM-RoBERTa-base (LoRA): The LoRA variant is markedly more stable than full FT on en-AU (spread 0.0162 vs. 0.1167), consistent with the parameter-efficient constraint acting as an implicit regulariser [6], [4]. All runs exceed 0.59 across all varieties, with en-UK and en-IN showing small spreads (0.0128 and 0.0119, respectively).

The zero spread of Qwen2.5 en-IN (all three seeds converging to exactly 0.6961) reflects three distinct optimisation paths reaching the same F1 ceiling, indicating a strong attractor corresponding to a shallow majority-bias-corrected solution; **a larger training budget would likely break this degeneracy**. The 800-sample training cap also limits variety-specific expressivity more broadly, particularly for en-IN; and the two XLM-R variants use different random classification-head seeds, which affects absolute gain figures but not the in-distribution LoRA-vs-full-FT comparison.

Variety	Validation Macro-F1			Spread (Max-Min)	Variety	Validation Macro-F1			Spread (Max-Min)
	Run 1	Run 2	Run 3			Run 1	Run 2	Run 3	
TinyLlama-1.1B (LoRA)					Qwen2.5-1.5B (LoRA)				
en-AU	0.7321	0.7366	**0.7465**	0.0144	en-AU	0.6648	**0.6888**	0.6806	0.0240
en-UK	**0.7301**	0.7098	0.7301	0.0203	en-UK	0.6380	**0.7285**	0.6930	0.0905
en-IN	0.6961	0.6795	**0.7317**	0.0522	en-IN	**0.6961**	0.6961	0.6961	0.0000
combined	0.7107	**0.7208**	0.7170	0.0101	combined	**0.7246**	0.7107	0.7084	0.0162
XLM-RoBERTa-base (Full FT)					XLM-RoBERTa-base (LoRA)				
en-AU	0.5859	**0.7026**	0.6937	0.1167	en-AU	0.7279	0.7217	**0.7379**	0.0162
en-UK	**0.6578**	0.6026	0.5999	0.0579	en-UK	0.6551	**0.6639**	0.6511	0.0128
en-IN	**0.6136**	0.5871	0.6011	0.0265	en-IN	**0.6052**	0.5933	0.5974	0.0119
combined	0.6013	0.6006	**0.6034**	0.0028	combined	**0.6614**	0.6476	0.6566	0.0138

Table 11: Comparison of stability (minimal-spread) across three training runs per variety for TinyLlama-1.1B (LoRA), Qwen2.5-1.5B (LoRA), XLM-RoBERTa-base (Full FT) and XLM-RoBERTa-base (LoRA) using validation. Bold=selected best checkpoint for the following results.

3.3.1.2 LoRA Experiment: In-Distribution Test Performance

This subsection reports per-class test metrics so that majority-class collapse is detectable [9], [10]. No model in this study collapses to a degenerate majority-class predictor: the lowest precision-sarcastic recorded is 0.1573 (XLM-R full-FT, en-IN) and the lowest recall-sarcastic is 0.1786 (TinyLlama, en-IN), both well above zero across all four architectures.

TinyLlama reaches its highest Macro-F1 on en-AU (0.7298) and lowest on en-IN (0.6057, Sarc-F1 0.2500, Recall-Sarc 0.1786, 82% of sarcastic en-IN instances missed despite 0.9265 accuracy, illustrating why **accuracy alone is misleading on this dataset** [1]). The combined adapter slightly beats the per-variety adapter on en-IN, the only variety where pooling helps for this model.

Qwen2.5 trails TinyLlama on en-AU and en-UK but marginally leads on en-IN (0.6218 vs 0.6057). Its zero-shot Sarcasm-F1 is markedly lower than TinyLlama's (e.g. 0.1014 vs 0.1672 on en-UK), reflecting a less-calibrated random head on this backbone, but **despite the weaker starting point LoRA closes most of the gap** to TinyLlama on en-AU, en-UK and combined and overtakes it on en-IN; its combined adapter outperforms the per-variety en-UK adapter by Δ=+0.0621, likely because the high en-UK run-to-run instability (Subsection 3.3.1.1) is smoothed by training on the larger pooled dataset.

XLM-R Full-FT over-predicts sarcasm (Recall-Sarc up to 0.8520 with low Precision-Sarc 0.16-0.47), **the opposite failure mode to the decoders**; its zero-shot collapses to F1-NonSarc≈0 because the random head predicts everything as sarcastic, so the resulting fine-tuning gains are best read as recovery from a pathological starting point rather than as evidence of backbone quality.

XLM-R LoRA improves every cell over Full FT (+0.05 to +0.07 Macro-F1) **and partially limits the over-prediction** (the Precision-Sarc / Recall-Sarc gap shrinks); the head-to-head ablation in Subsection 3.3.2 isolates this effect. Its zero-shot rows show the opposite collapse mode, F1-Sarc=0 across all varieties because that random head seed predicts everything as non-sarcastic. The two opposite collapse modes (everything-sarcastic vs everything-non-sarcastic) come from different random

classification-head seeds and illustrate why **zero-shot baselines for randomly-headed sequence classification are dominated by initialisation luck**; all four fine-tuned conditions clear them substantially.
On en-AU, where TinyLlama performs best, it reaches Sarcasm-F1 0.6414, within 0.04 of Srirag et al.'s variety-averaged 0.68 for Mistral-22B [1], despite being a 20× smaller model with ~0.1% trainable parameters. Our XLM-R-base LoRA reaches 0.6433 on en-AU, surpassing the 0.62 scored by their XLM-R-Large full fine-tuning [1], a model twice the size, suggesting that **parameter-efficient adaptation on a smaller backbone is competitive with full fine-tuning on a larger one**. The encoder-beats-decoder pattern also replicates their core finding on this parameter-efficient, small-data subset (see Figure 29 in Appendix).

Model	Contndition	In-Distribution (Train-Eval Variety is same)	Acc	Macro-F1	F1-Sarc	P-Sarc	R-Sarc	F1-Non-Sarc	P-Non-Sarc	R-Non-Sarc
TinyLlama-1.1B (LoRA)	Zero-shot	en-AU	0.6042	0.5492	0.3917	0.3571	0.4337	0.7067	0.7413	0.6752
	LoRA		0.7586	**0.7298**	**0.6414**	0.5692	0.7347	0.8181	0.8744	0.7686
	Zero-shot	en-UK	0.6014	0.4526	0.1672	0.0993	0.5283	0.7380	0.9402	0.6074
	LoRA		0.9114	**0.7131**	**0.4746**	0.4308	0.5283	0.9516	0.9606	0.9428
	Zero-shot	en-IN	0.7733	0.5231	0.1778	0.1183	0.3571	0.8685	0.9444	0.8039
	LoRA		0.9265	**0.6057**	**0.2500**	0.4167	0.1786	0.9613	0.9419	0.9816
	Zero-shot	combined	0.6670	0.5262	0.2679	0.1933	0.4361	0.7845	0.8849	0.7045
	LoRA		0.8585	**0.7168**	**0.5164**	0.4940	0.5410	0.9171	0.9243	0.9100
Qwen2.5-1.5B (LoRA)	Zero-shot	en-AU	0.5172	0.4405	0.2333	0.2188	0.2500	0.6477	0.6682	0.6285
	LoRA		0.7436	**0.7172**	**0.6307**	0.5468	0.7449	0.8037	0.8750	0.7431
	Zero-shot	en-UK	0.6457	0.4404	0.1014	0.0628	0.2642	0.7794	0.9182	0.6770
	LoRA		0.9214	**0.6396**	**0.3210**	0.4643	0.2453	0.9583	0.9405	0.9768
	Zero-shot	en-IN	0.4265	0.3617	0.1583	0.0880	0.7857	0.5651	0.9620	0.4000
	LoRA		0.9203	**0.6218**	**0.2857**	0.3714	0.2321	0.9578	0.9449	0.9711
	Zero-shot	combined	0.5245	0.4188	0.1709	0.1130	0.3508	0.6667	0.8398	0.5527
	LoRA		0.8429	**0.6965**	**0.4858**	0.4475	0.5311	0.9073	0.9215	0.8935
XLM-RoBERTa-base (Full FT)	Zero-shot	en-AU	0.2939	0.2271	0.4542	0.2939	1.0000	0.0000	0.0000	0.0000
	Full FT		0.6687	**0.6591**	**0.6018**	0.4652	0.8520	0.7163	0.9058	0.5924
	Zero-shot	en-UK	0.0757	0.0704	0.1408	0.0757	1.0000	0.0000	0.0000	0.0000
	Full FT		0.7771	**0.6103**	**0.3554**	0.2275	0.8113	0.8653	0.9804	0.7743
	Zero-shot	en-IN	0.0686	0.0642	0.1284	0.0686	1.0000	0.0000	0.0000	0.0000
	Full FT		0.7230	**0.5432**	**0.2566**	0.1573	0.6964	0.8298	0.9701	0.7250
	Zero-shot	combined	0.1397	0.1226	0.2452	0.1397	1.0000	0.0000	0.0000	0.0000
	Full FT		0.6803	**0.6014**	**0.4241**	0.2834	0.8426	0.7787	0.9624	0.6539
XLM-RoBERTa-base (LoRA)	Zero-shot	en-AU	0.7046	0.4134	0.0000	0.0000	0.0000	0.8267	0.7057	0.9979
	LoRA		0.7556	**0.7287**	**0.6433**	0.5632	0.7500	0.8141	0.8793	0.7580
	Zero-shot	en-UK	0.9243	0.4803	0.0000	0.0000	0.0000	0.9607	0.9243	1.0000
	LoRA		0.8286	**0.6639**	**0.4286**	0.2866	0.8491	0.8992	0.9853	0.8269
	Zero-shot	en-IN	0.9289	0.4816	0.0000	0.0000	0.0000	0.9632	0.9312	0.9974
	LoRA		0.8088	**0.6052**	**0.3217**	0.2126	0.6607	0.8887	0.9704	0.8197
	Zero-shot	combined	0.8589	0.4621	0.0000	0.0000	0.0000	0.9241	0.8601	0.9984
	LoRA		0.7714	**0.6599**	**0.4652**	0.3455	0.7115	0.8546	0.9434	0.7812

Table 12: Per-class test metrics for all four architectures on every in-distribution test condition, with zero-shot baselines The Train-Eval column shows the variety used for both training and evaluation; for the frozen-backbone zero-shot rows this column lists only the evaluation variety because no training was performed. Every fine-tuned row is an in-distribution condition where Train = Eval, corresponding to the diagonal cells of the 4x4 cross-variety matrices. Bold = in-distribution fine-tuned Macro-F1 and Sarcasm-F1. Columns: P-NonSarc/R-NonSarc = precision/recall on the non-sarcastic class; P-Sarc/R-Sarc = precision/recall on the sarcastic class.

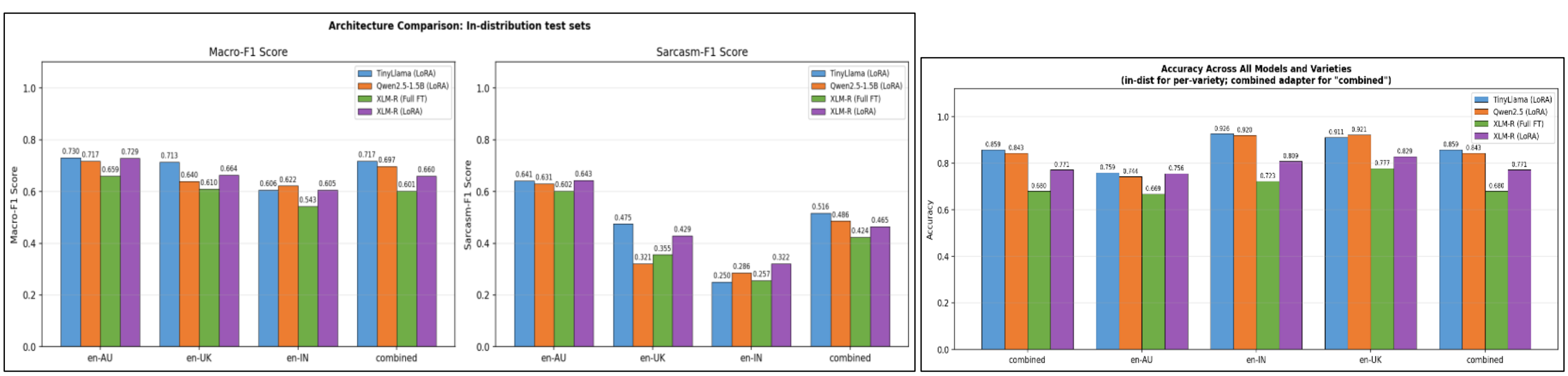

Figure 25: In-distribution comparison of Zero-Shot vs Per-Variety LoRA vs Combined LoRA test metrics per variety for TinyLlama-1.1B (LoRA), Qwen2.5-1.5B (LoRA), XLM-RoBERTa-base (Full FT) and XLM-RoBERTa-base (LoRA). Zero-shot (frozen model, random head). XLM-RoBERTa-base (LoRA) shows Sarcasm-F1 of zero (0) because the randomly initialised classification head in that specific xlmr_lora zero-shot run predicted every single test instance as Non-Sarcastic.

3.3.1.3 LoRA Experiment: Adaptation Gain Analysis

This subsection quantifies two things per variety: the marginal benefit of fine-tuning each model over its own zero-shot baseline (the per-variety LoRA / Full-FT gain) and whether the pooled combined adapter substitutes for, exceeds, or underperforms variety-specific training (the Winner column).

Decoder LLMs: TinyLlama gains the most on en-UK and en-AU and the least on en-IN, a differential explained not by pre-training coverage but by the lower volume and prevalence of sarcastic en-IN training instances [1], [9]. The reversal where the combined adapter outperforms per-variety on en-IN shows that the 800-sample budget x≈13% en-IN sarcasm prevalence is too thin for a highly specific en-IN adapter and that exposure to richer en-AU (≈42%) and en-UK (≈22%) signal during combined training provides complementary learning. Qwen2.5 shows larger gains across all varieties driven by lower zero-shot starting points, with the combined adapter winning over per-variety on both en-UK and en-IN.

XLM-R variants: Full FT shows the largest absolute gains in the study, but only because its zero-shot starting point is catastrophically low, the random head collapses to predicting almost everything as sarcastic, so these gains read as recovery from a pathological baseline rather than evidence of backbone quality. The XLM-R LoRA variant has a higher zero-shot baseline under a different random head seed (here the head collapses to all-non-sarcastic), so its smaller headline gains understate model quality. The two XLM-R zero-shot baselines used different random head seeds and are not directly comparable; the fair LoRA-vs-FT comparison is the in-distribution head-to-head Subsection 3.3.2.

Aggregating the Winner column across all four architectures reveals a structural asymmetry in pooling behaviour. Decoder LoRA models (TinyLlama, Qwen2.5) benefit from the combined adapter specifically on en-IN, where the 800-sample training budget combined with ~13% sarcasm prevalence is too thin to support a specialised per-variety adapter. XLM-RoBERTa shows the opposite pattern: it prefers the combined adapter on en-AU but per-variety training on en-IN, indicating that the encoder specialises sharply enough on the outer-circle signal within 800 samples that pooling dilutes rather than enriches its representation. **Pooling (combining adapters) is therefore not universally beneficial; the optimal choice is architecture- and variety-specific**.

Model	Variety	*Macro-F1*			Gain		Winner
		Zero-Shot	In-Dist & LoRA	Comb. & LoRA	In-Dist & LoRA	Comb. & LoRA	
TinyLlama-1.1B (LoRA)	en-AU	0.5492	0.7298	0.7242	+0.1806	+0.1750	Per-Variety
	en-UK	0.4526	0.7131	0.7082	+0.2605	**+0.2556**	**Per-Variety**
	en-IN	0.5231	0.6057	0.6279	+0.0826	+0.1048	Combined
	combined	0.5262	-	0.7168	-	+0.1906	-
Qwen2.5-1.5B (LoRA)	en-AU	0.4405	0.7172	0.6929	+0.2767	+0.2524	Per-Variety
	en-UK	0.4404	0.6396	0.7017	+0.1992	+0.2613	Combined
	en-IN	0.3617	0.6218	0.6344	+0.2601	**+0.2727**	**Combined**
	combined	0.4188	-	0.6965	-	+0.2777	-
XLM-RoBERTa-base (Full FT)	en-AU	0.2271	0.6591	0.7024	+0.4320	+0.4753	Combined
	en-UK	0.0704	0.6103	0.5982	+0.5399	**+0.5278**	**Per-Variety**
	en-IN	0.0642	0.5432	0.4812	+0.4790	+0.4170	Per-Variety
	combined	0.1226	-	0.6014	-	+0.4788	-
XLM-RoBERTa-base (LoRA)	en-AU	0.4134	0.7287	0.7472	+0.3153	**+0.3338**	**Combined**
	en-UK	0.4803	0.6639	0.6523	+0.1836	+0.1720	Per-Variety
	en-IN	0.4816	0.6052	0.5508	+0.1236	+0.0692	Per-Variety
	combined	0.4621	-	0.6599	-	+0.1978	-

Table 13: TinyLlama-1.1B (LoRA), Qwen2.5-1.5B (LoRA), XLM-RoBERTa-base (Full FT) and XLM-RoBERTa-base (LoRA): LoRA gain over zero-shot per variety. Pooling (combining adapters) is not universally beneficial.

3.3.2 LoRA Experiment: Full Fine-Tuning vs. LoRA Ablation

LoRA improves the XLM-R-base baseline by +0.05 to +0.07 Macro-F1 and +0.04 to +0.07 Sarcasm-F1 on every in-distribution cell. This head-to-head comparison does not depend on zero-shot baselines and is therefore unaffected by the random head-initialisation seed asymmetry noted in Subsection 3.3.1.3, the only varying factor between the two columns is full-parameter update vs rank-8 LoRA update, with backbone weights, tokeniser, data splits, hyperparameters, optimiser, seeds (43/44/45) and hardware all held constant. The mechanism is visible in the training-stability results in Subsection 3.3.1.1: the en-AU spread is 0.1167 for Full FT versus 0.0162 for LoRA, **a 7.2x reduction in run-to-run variance**. Updating all 278 M parameters on only 800 samples exposes the optimisation to high initialisation variance, a sign of overfitting in the small-data regime. Restricting trainable weights to ≈0.3 M LoRA parameters (≈0.1% of the model) imposes a low-rank prior on the weight update $\Delta W=BA$, which acts as an implicit regulariser matching the near-low-rank structure of gradient updates predicted by matrix-completion theory [13] and by the original LoRA hypothesis [6]. This is consistent with broader parameter-efficient transfer-learning findings [4] and is only observable because both XLM-R conditions were included as a controlled ablation.

Variety	Full-FT Macro-F1	LoRA Macro-F1	Δ Macro-F1	Full-FT Sarc-F1	LoRA Sarc-F1	Δ Sarc-F1
en-AU	0.6591	0.7287	**+0.0696**	0.6018	0.6433	**+0.0415**
en-UK	0.6103	0.6639	**+0.0536**	0.3554	0.4286	**+0.0732**

en-IN	0.5432	0.6052	**+0.0620**	0.2566	0.3217	**+0.0651**
combined	0.6014	0.6599	**+0.0585**	0.4241	0.4652	**+0.0411**

Table 14: XLM-RoBERTa-base: full fine-tuning vs LoRA on every in-distribution test cell. LoRA wins every cell on both Macro-F1 and Sarcasm-F1. Absolute Macro-F1 values are higher than Full FT on every cell.

Overall, **LoRA beats full fine-tuning on the same backbone (XLM-R-Base)** by 0.05-0.07 Macro-F1 and 0.04-0.07 Sarcasm-F1 **on every in-distribution cell**, attributable to the implicit **low-rank regularisation** that **mitigates overfitting** when all 278M parameters are updated on only 800 balanced samples.

3.3.3 LoRA Experiment: Cross-Variety Transferability Analysis

3.3.3.1 LoRA Experiment: Per-Variety Transfer Matrix

Each best adapter is evaluated on every test set, forming a 4x4 matrix (3 per-variety adapters + 1 combined adapter x 4 test sets). Diagonal cells are in-distribution performance; off-diagonal cells are cross-variety transfer. The variety gap per variety is the in-distribution Macro-F1 minus the mean of the two per-variety off-diagonal cells. Across the four matrices, four cross-architecture patterns are visible:

TinyLlama: the en-IN adapter transfers worst to en-AU (0.4609), the worst off-diagonal cell, reflecting **lexical and pragmatic distance between outer-circle Indian English and colloquial Australian English** [2]; the en-AU ↔ en-UK inner-circle pair shows near-symmetric transfer.

Qwen2.5: en-AU → en-IN transfer (0.4622) **is worse than TinyLlama's** (0.5156) **despite multilingual pre-training**, and the **combined adapter dramatically outperforms per-variety en-UK** (0.7017 vs 0.6396).

XLM-R Full FT: the **en-UK → en-AU off-diagonal** (0.6609) **actually exceeds en-UK in-distribution** (0.6103); the **en-AU adapter transfers substantially better to en-UK** (0.5577) **than to en-IN** (0.4389), a 0.1188 Macro-F1 gap that quantifies the inner-circle → outer-circle boundary predicted by Kachru [2].

XLM-R LoRA: the qualitative pattern mirrors Full FT, but absolute scores are uniformly higher; the en-IN LoRA adapter transfers to en-UK at 0.6532, higher than its own in-distribution score of 0.6052, reproducing the negative-variety-gap phenomenon that becomes the central finding of the next subsection (see Figure 32 and Figure 31 in Appendix).

Macro-F1									
TinyLlama-1.1B (LoRA)					**Qwen2.5-1.5B (LoRA)**				
Train↓\Test→	en-AU	en-UK	en-IN	combined	Train↓\Test→	en-AU	en-UK	en-IN	combined
en-AU	**0.7298**	0.5870	0.5156	0.6215	en-AU	**0.7172**	0.5535	0.4622	0.5882
en-UK	0.5658	**0.7131**	0.6078	0.6216	en-UK	0.5274	**0.6396**	0.5917	0.5817
en-IN	0.4609	0.5827	**0.6057**	0.5365	en-IN	0.4230	0.5563	**0.6218**	0.5192
combined	0.7242	0.7082	0.6279	**0.7168**	combined	0.6929	0.7017	0.6344	**0.6965**
XLM-RoBERTa-base (Full FT)					**XLM-RoBERTa-base (LoRA)**				
Train↓\Test→	en-AU	en-UK	en-IN	combined	Train↓\Test→	en-AU	en-UK	en-IN	combined
en-AU	**0.6591**	0.5577	0.4389	0.5611	en-AU	**0.7287**	0.6055	0.4909	0.6158
en-UK	0.6609	**0.6103**	0.4843	0.5966	en-UK	0.6800	**0.6639**	0.5155	0.6215
en-IN	0.5983	0.6366	**0.5432**	0.5930	en-IN	0.5683	0.6532	**0.6052**	0.6066
combined	**0.7024**	0.5982	0.4812	0.6014	combined	**0.7472**	0.6523	0.5508	**0.6599**

Table 15: 4x4 Macro-F1 cross-variety matrix comparison per variety for TinyLlama-1.1B (LoRA), Qwen2.5-1.5B (LoRA), XLM-RoBERTa base (Full FT) and XLM-RoBERTa-base (LoRA) using validation. Bold=selected best checkpoint for the following results.

3.3.3.2 LoRA Experiment: Comparative Variety Gap Summary

The table below consolidates the variety gap across all four architectures. The decoder LoRA architectures show mean gaps of 0.13-0.14 on the per-variety rows, whereas **both XLM-RoBERTa variants show substantially smaller mean gaps (0.04-0.08)**. Both XLM-R variants exhibit a near-zero or negative variety gap on en-IN (-0.0743 for full FT, -0.0056 for LoRA), meaning the en-IN fine-tuned encoder transfers as well or better to other varieties than it performs in-distribution. This reversal, absent in both decoder architectures, demonstrates that **XLM-RoBERTa's bidirectional representations [12] capture more transferable cross-variety pragmatic features**, a concrete finding for multi-dialect deployment scenarios. A complementary interpretation is that the en-IN test split may itself contain noisier or harder labels than the inner-circle splits, outer-circle minority varieties are known to attract higher annotation disagreement [10], which would partially account for why an en-IN-trained adapter scores higher on en-AU/en-UK test cells than on its own in-distribution cell.

The bottom row reports the same metric for the combined adapter, with the diagonal cell being combined → combined and the off-diagonal cells being combined → en-AU, combined → en-UK and combined → en-IN. These gaps are uniformly small (0.0075-0.0300) across all four architectures - unsurprising because the combined adapter was trained on the union of the three varieties, so none of the off-diagonal test sets is strictly out-of-domain. **The combined row therefore measures within-pool consistency rather than out-of-pool transfer**, and should not be directly compared to the per-variety rows above it.

Model	en-AU Gap	en-UK Gap	en-IN Gap	Per-variety Mean	Combined Gap
TinyLlama-1.1B (LoRA)	0.1785	0.1263	0.0839	0.1296	0.0300
Qwen2.5-1.5B (LoRA)	0.2093	0.0800	0.1321	0.1405	0.0202
XLM-RoBERTa-base (Full FT)	0.1608	0.0377	**-0.0743**	**0.0414**	0.0075
XLM-RoBERTa-base (LoRA)	0.1805	0.0662	**-0.0056**	0.0803	0.0098

Table 16: Variety gap = how much an adapter's Macro-F1 drops on other varieties compared to its own in-distribution test set. For the first three columns (en-AU / en-UK / en-IN Gap), this is the adapter's diagonal score minus the average of its scores on the two other per-variety test sets, a measure of out-of-distribution transfer. For the Combined Gap column, the diagonal is the combined -> combined cell, and the off-diagonal is the adapter's average score on the three per-variety test sets; because the combined adapter was trained on a pool of all three varieties, none of those test sets is strictly out-of-distribution, so the Combined Gap measures within-pool consistency rather than transfer and is not directly comparable to the per-variety columns.

In this small-data regime (800 samples, 1.1B-1.5B decoder parameters), both XLM-RoBERTa variants show near-zero or negative variety gaps specifically on en-IN (-0.07 and -0.01 respectively), whereas decoder gaps on the same variety reach 0.08-0.13, suggesting that **bidirectional encoders offer more robust cross-variety transfer for outer-circle English under constrained resources**. This is consistent with Srirag et al. [1], who find that encoder models generally outperform decoders on BESSTIE, attributing the gap to encoders being structurally better suited to sequence classification tasks.

4. Sarcasm Explanation & Error Analysis

4.1 Methodology and Error Selection

The aim of this section is to look at the failure modes of the best LoRA adapter from Section 2.3 and to see if a four-shot prompt with linguistic explanations can fix errors that the adapter alone cannot. The analysis is limited to in-distribution errors. Each adapter is tested on the test split of the variety it was trained on, so we can keep pragmatic difficulty separate from any loss from cross-dialect transfer. We use the TinyLlama 1.1B LoRA family [7] as the source of errors because it gets the lowest base model perplexity on all three varieties (Table 7). This rules out a calibration reason for the residual mistakes and lets the analysis focus on the pragmatic sarcasm signal.

From the set of errors across the three TinyLlama adapters, we randomly picked ten examples. They cover both error types, five False Negatives (sarcastic posts that were predicted as non-sarcastic) and five False Positives (non-sarcastic posts that were predicted as sarcastic). These selections covered all three varieties. We randomly picked samples that we could explain in linguistic terms. Each example had to allow an explanation of its true label using standard sarcasm mechanisms [3], not just specific context.

4.2 Explanation Design

We picked four examples from the **exports from the best LLM, TinyLlama1.1B with LoRA applied, on the sarcasm task** to analyse. The set is balanced by label, with two sarcastic and two non-sarcastic, on purpose. This stops a steady label trend from forming at the end of the few-shot context. Examples 3 (en-AU) and 5 (en-IN) show two sarcasm subtypes. The first is a positive adjective placed next to a negative real situation ("It's great, barely any customers and the cinemas are always empty"). The second is fake enthusiasm about a bad list ("Will we die of heat, dehydration or asphyxiation? So much to choose from!"). Examples 2 (en-UK) and 7 (en-UK) show the opposite failure mode, which is strong emotional language without the meaning flip that sarcasm needs. The word "Brilliant" in "I laughed out loud at that. Brilliant" is a plain positive reaction, and "ruined everything" in "The internet has ruined everything" is a sincere, over-the-top complaint rather than an ironic flip.

Each explanation follows a three-step structure based on the metacognitive prompting method of Singh et al. [11]. First, find the surface polarity of the words. Second, compare it with the real situation the post is talking about. Third, decide whether there is a meaning flip. The four examples are ordered Sarcastic, Sarcastic, Non Sarcastic, Non Sarcastic, so that the two explanations closest to the test input are non sarcastic. This helps with the recency bias that has been noted in small decoder language models, where the last in-context examples have too much weight on the prediction for the next input.

4.3 Prompt Templates

We built two prompts to isolate the effect of the explained examples. The zero-shot prompt has only a dialect-aware task description and the test input. The four-shot prompt is the same as the zero-shot one, but with the four explanations added before the test input. Both templates end with the chain of thought trigger "Let's work this out in a step-by-step way to be sure we have the right answer" [17], followed by the prefix "This text is ". The trigger pushes the model to reason in steps, and the prefix locks the next token to a classification answer rather than a new example. Inference uses greedy decoding (do_sample=False) to remove sampling variance. To pull the prediction out of the generation, we use a two-stage parser. The first rule looks for an "Answer:" line in the output. The fallback handles negation by scanning for the last "sarcastic" or "not sarcastic" in the text, which catches generations that do not follow the expected answer format.

A note on scope. We evaluate the zero-shot and four-shot prompts on the base TinyLlama 1.1B Chat model with no LoRA weights. This separates the effect of prompting from the effect of the adapter and gives a clean three-way comparison. The first is the LoRA adapter prediction from Section 2.3. The second is the base model with zero-shot prompting. The third is the base model with four-shot prompting and explanations.

In terms of structure, the four-shot prompt follows a fixed template. It starts with a system instruction that tells the model it is a sarcasm detector for varieties of English. Then come four labelled examples. Each one has the input text, the English variety, a step-by-step reasoning explanation, and the true answer. The four examples end with "=== END OF EXAMPLES ===" before the new test input. A full version of both prompts is in the submitted notebook.

4.4 Results

Table 17 shows the three-way comparison across the six held-out test examples, and Figure 26 shows the recovery rate by error type. The LoRA adapter is wrong on every row by design, since these are the errors we are probing. Zero shot prompting of the base TinyLlama 1.1B Chat model recovers three out of six errors (50.0%), and four shot prompting recovers four out of six (66.7%). That is a net gain of one example over zero shot.

Variety	Text (truncated)	True Label	LoRA Adapter	Zero-Shot	Four-Shot
en-UK	Its great we cant defend ourselves. I really hope all four...	Sarcastic	Non-Sarc	Sarc ✓	Sarc ✓
en-AU	Rich families that own houses getting a free reno. Just love...	Sarcastic	Non-Sarc	Sarc ✓	Sarc ✓
en-AU	so you're a genius and a mind reader WOW.	Sarcastic	Non-Sarc	Sarc ✓	Sarc ✓
en-UK	Dystopian nightmare pod.	Non-Sarcastic	Sarc	Sarc ✗	Non-Sarc ✓
en-UK	Boris.... always the bumbly fumbly 'bants' attention whore...	Non-Sarcastic	Sarc	Sarc ✗	Sarc ✗
en-IN	I'm with you people, DO NOT LET HIM GET AWAY	Non-Sarcastic	Sarc	Sarc ✗	Sarc ✗

Table 17: Three way comparison of predictions on the six held out test examples taken from the errors of the TinyLlama 1.1B LoRA adapters. A check mark means the prediction matches the true label, and a cross means it does not. The LoRA adapter is wrong on every row by design.

The breakdown by error type is revealing. Zero-shot prompting recovers all three False Negatives but none of the three False Positives, because every zero-shot prediction is "Sarcastic," matching the true label on FNs by chance rather than through real discrimination. Four-shot prompting keeps perfect sarcastic recall (3/3) while additionally recovering one False Positive, example 8 ("Dystopian nightmare pod"), bringing non-sarcastic recall from 0/3 to 1/3.

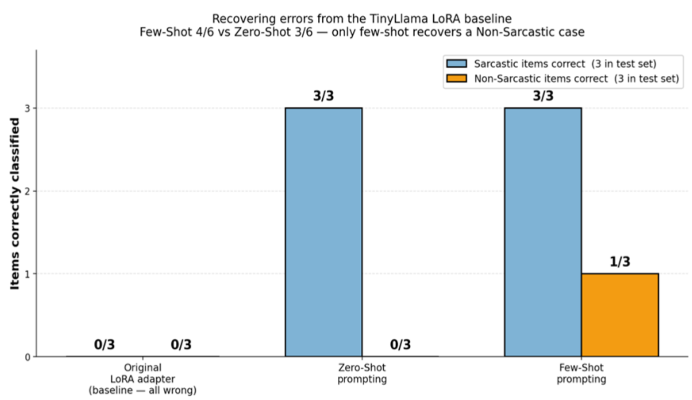

Figure 26: Recovery rate for six TinyLlama 1.1B LoRA adapter errors under three settings. The first is the original adapter. The second is zero shot prompting of the base model. The third is four shot prompting with linguistic explanations. Four shot recovers one

The raw outputs show a qualitative difference. Under zero-shot, example 8 receives shallow feature listing ("The text is in the past tense. The text is about a pod"), surface-level points that never check whether the speaker means the opposite of what they say. Under four-shot, the model follows the three-step reasoning template, first determining dialect, then checking the link between word polarity and real intent. Both setups initially generate "100% sarcasm," but the chain-of-thought reasoning in the four-shot setup leads the model to revise to Non-Sarcastic, matching the true label. This confirms that the explained examples change how the model reasons, not just which label it favours.

4.5 Discussion and Conclusion

The recovered False Positive (example 8) shares a clear structural pattern with example 7 ("The internet has ruined everything"): both are short dramatic phrases expressing real negative feeling without flipping meaning. Example 2 ("I laughed out loud at that. Brilliant") teaches the matching rule but shares less structure with example 8. Examples 9 and 10, which are not recovered, are longer and more conversational, suggesting explanations transfer best when the test item shares the syntactic register of the explained examples, though with only six test items this could also be the prompt reducing the model's tendency to predict 'Sarcastic' when input resembles the non-sarcastic examples. Across per-variety slices (en-AU 2/2 in both setups, en-UK 1/3 → 2/3, en-IN 0/1 in both), the en-UK gain lines up with en-UK being best-represented in the explained set (two of four examples). With only one en-IN test item, it is unclear whether this is a real cross-dialect effect or coincidence.

Overall, four-shot prompting gives a modest gain over zero-shot (4/6 vs 3/6), coming only from recovering one False Positive that structurally matches the explained examples. Zero-shot cannot address False Positives on this backbone, as the base model predicts sarcasm for every input, a label bias the few-shot examples only partly correct. Brown et al. [15] and Wei et al. [16] showed that few-shot and chain-of-thought performance depends heavily on model scale; TinyLlama at 1.1B is well below those thresholds, placing a ceiling on prompt engineering alone. Nonetheless, explanation-based prompting is worth doing even on a small model, as long as explanations match the structure of expected failure modes. For production deployment, a larger base model or class-weighted fine-tuning would help more than prompt design alone on a 1.1B-parameter backbone.

5. Deployment Endpoint

Now that the architectures have been developed, the next objective is to build a web service to host the best-performing trained version of each model for each English variety separately (British (en-UK), Australian English (en-AU), and Indian English (en-IN)). To demonstrate the variety of developed architectures, a web-based deployment endpoint was created using the Gradio framework. The web service has been developed to allow users to select an architecture, choose a language variety, and input a comment before running the inference. The output is presented as either sarcastic or non-sarcastic, alongside a confidence score (%) and the inference time for each selected model run. In addition to supporting the primary objective of the sarcasm detection system, there is a flagging option that allows users to signal potentially incorrect results for error analysis.

5.1 Interface Design

5.1.1 Framework Selection: Gradio Compared with Alternative Frameworks

Gradio was chosen over Flask and Streamlit for its strong suitability for rapid deployment of ML inference interfaces with high-level interactive components that work directly with Python functions [26]. Flask would require separate frontend development and manual integration of model controls. Streamlit, while effective for dashboards, introduces unnecessary overhead through its reactive rerun model when users frequently switch between model architectures and LoRA adapters [27]. Gradio's callback-driven design handles this naturally, maintaining stability across architectures while providing built-in support for runtime feedback, performance monitoring, and user flagging in a standalone Python script.

5.1.2 User Interface Architecture and Runtime Execution

A key design decision was to separate each deployed model into its own tab, giving clear organisation and easier comparative performance analysis. The models were organised into three main categories: (1) Baseline Models: TF-IDF, RoBERTa, and FastText, (2) Cross-Variety Encoders: DeBERTa, RemBERT, and XLM-RoBERTa, (3) LLM Variation Models: TinyLlama, XLMR, Qwen2.5, and XLMR+LoRA. For architectures where the task is split between two specialised models, the interface displays both the sarcasm classifier and sentiment classifier models simultaneously. The sarcasm classifier will display the outputs as either Sarcastic or Non-Sarcastic. The sentiment classifier outputs either Negative or Positive, where Negative indicates sarcasm/not sentimental. This allows the user to compare both results simultaneously. Below the results section, a flag button allows users to store any incorrect results automatically in a specific CSV file separated by model type (e.g. flagged_tinyllama_results.csv). Each file stores the model name, language variety, input comment, result, and the confidence score for future retraining and error analysis.

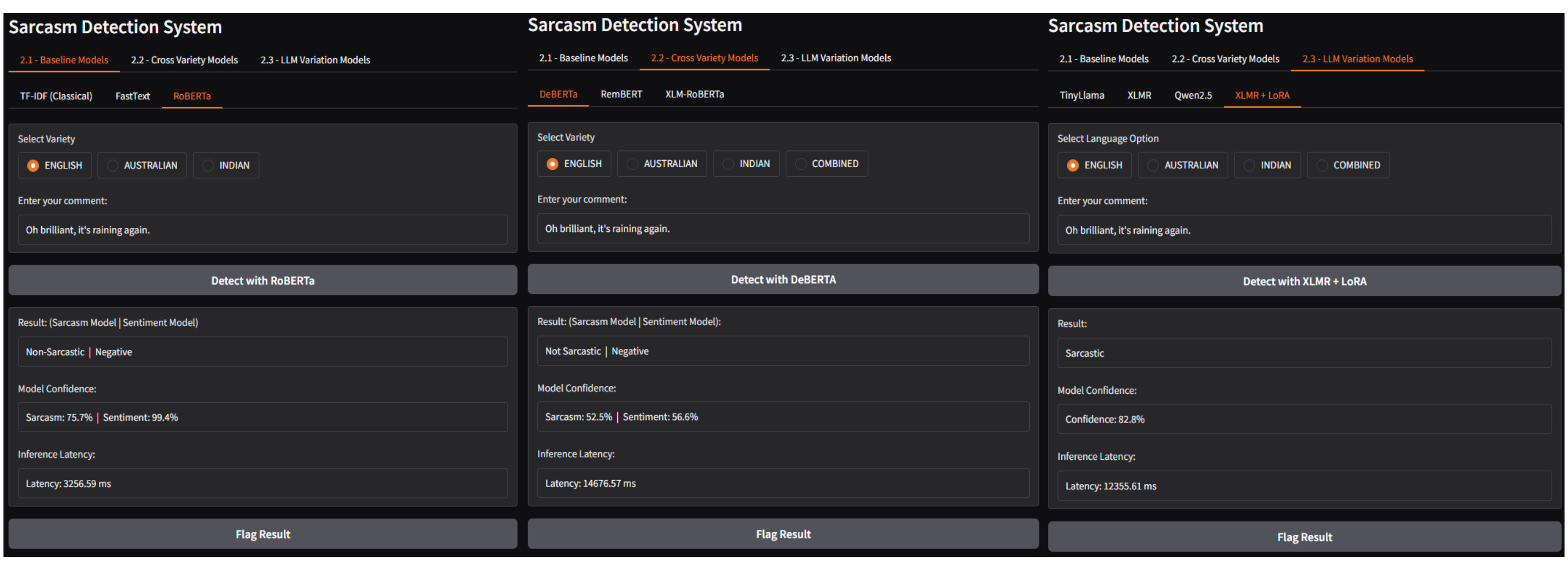

Figure 27: Three Examples of the deployed Gradio Interface Showing Model Tabs, Text Input, Prediction Output, Confidence Score, Inference Time, and Flagged Controls. Here the same sarcastic phrase is compared across models in the same variety for fair comparison of performance.

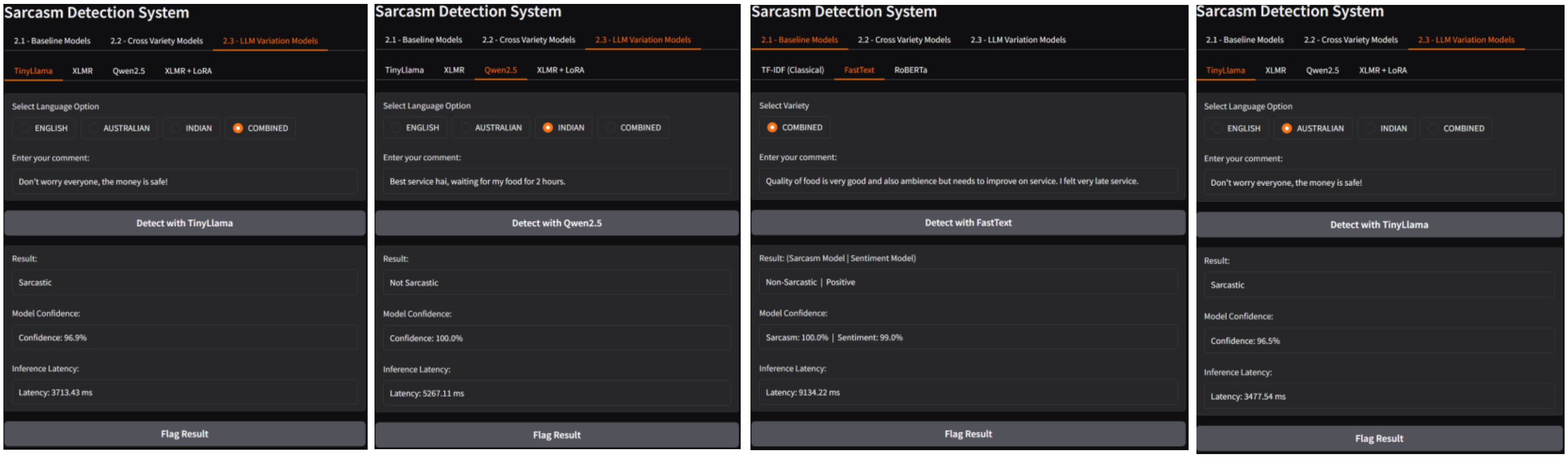

Figure 28: Four Examples of the deployed Gradio Interface Showing Model Tabs, Text Input, Prediction Output, Confidence Score, Inference Time, and Flagged Controls. Here a mixture of different sarcastic and positive texts were tested across different models and different varieties to show the completeness of the UI and the robustness of the backend.

5.1.3 Backend Architecture and Model Routing

The backend system supports inference across multiple models through a consistent deployment pipeline. User selections made through the interface are passed to the backend, which loads the relevant trained models and their adapters stored within the inference pipeline, processes the input text, and returns the predicted result, confidence score, and inference time. Inference wrappers abstract over different model types so that all architectures can be served through a unified interface. To manage deployment across English language varieties, the system uses a registry-based architecture: a centralised mapping directs the user's variety selection to the relevant adapter path or model checkpoint, allowing variety-specific models to be loaded without retraining. This design minimises redundant code, replacing unique functions per variety with a single registry lookup, and supports future scalability and availability, as new varieties or architectures can be added without re-engineering the core inference pipeline. All model checkpoints and LoRA adapters are stored locally within the project directory. At inference time, an ADAPTER_REGISTRY dictionary maps each architecture-variety pair (e.g., tinyllama × en-AU) to its adapter subdirectory. The Gradio radio button selection is converted via a UI mapping (e.g., "AUSTRALIAN" → en-AU) into a registry key, which resolves the full adapter path passed to load_model_and_predict(). An identical pattern applies to Section 2.1 models via a separate MODEL_REGISTRY keyed by architecture, task, and variety. This registry design means adding a new variety or model requires a single dictionary entry rather than new inference code. For submission portability, adapters could alternatively be hosted on HuggingFace Hub and loaded via from_pretrained(), avoiding large files in the ZIP.

5.2 Efficiency Analysis

Efficient inference is a practical concern for any NLP system that needs to run in a real deployment setting, where response time and resource constraints directly affect how useful the system actually is. This section evaluates inference latency across all model families developed throughout this coursework, covering classical baselines and transformer models from Section 2.1, cross-variety encoder models from Section 2.2, and LLM variations from Section 2.3. The goal is to understand the speed versus performance trade-off across architectures of varying complexity and size, and to provide practical guidance on which model best suits different deployment scenarios.

5.2.1 Classical and Transformer Baseline Inference

Inference time was measured for all Section 2.1 models using CUDA synchronisation where applicable, with results averaged across 3 seeds. Standard inference was measured on the full test set of 2,183 samples, and large batch experiments were conducted at batch sizes of 100, 500, and 2,183 samples to assess how each model scales with increasing input size.

5.2.1.1 Standard Inference

Model	Task	Per-Sample Mean (ms)	Per-Sample Std (ms)	Macro-F1 Mean	Macro-F1 (Std)
FastText	Sarcasm	0.0859	0.0272	0.5957	0.0226
	Sentiment	0.1331	0.1348	0.8066	0.0295
LinearSVM	Sarcasm	0.4425	0.3938	0.5932	0.0
	Sentiment	0.2227	0.0525	0.8407	0.0
LogisticRegression	Sarcasm	0.2623	0.0432	0.6275	0.0
	Sentiment	0.2729	0.0253	0.8287	0.0
RoBERTa	Sarcasm	2.3807	0.0248	0.7004	0.0117

	Sentiment	2.3193	0.1194	0.8915	0.0066

Table 18: Standard inference time and Macro-F1 for Section 2.1 models (mean ± std across 3 seeds, 2,183 test samples).

Table 18 reports mean per-sample inference time and Macro-F1 for each model on both Sentiment and Sarcasm tasks. FastText is the fastest model overall at 0.086 ms per sample on the Sarcasm task, which is expected given its shallow architecture that relies on fixed word embeddings without any attention computation [18]. LogisticRegression and LinearSVM follow at 0.26 ms and 0.44 ms respectively, both operating on sparse TF-IDF vectors with no learned representations. RoBERTa is considerably slower at 2.38 ms per sample, which reflects the cost of running a 12-layer transformer where self-attention scales quadratically with sequence length [14]. Despite this overhead, RoBERTa achieves a Sarcasm Macro-F1 of 0.7004 compared to 0.5957 for FastText and 0.6275 for LogisticRegression, a meaningful gain that justifies the extra latency in settings where prediction quality matters more than raw speed [5]. It is also worth noting that LinearSVM and LogisticRegression show zero standard deviation across all three seeds. This is expected behaviour rather than a bug, since both algorithms are fully deterministic given fixed dataset splits and TF-IDF vectors. FastText and RoBERTa show small but non-zero variance, which reflects the effect of random weight initialisation during neural network training.

5.2.1.2 Large Batch Inference

To assess how models scale under heavier load, LogisticRegression and RoBERTa were tested at batch sizes of 100, 500, and 2,183 samples. Results are shown in Table 19.

LogisticRegression scales near-linearly with batch size, with per-sample time decreasing slightly from 0.037 ms at batch 100 to 0.026 ms at the full test set size. This modest improvement comes from better CPU cache utilisation as the vectorised matrix multiplication in scikit-learn becomes more efficient over larger inputs.

RoBERTa shows a more interesting pattern worth discussing. At batch size 100, the per-sample time is 3.83 ms, which is actually higher than the standard test set mean of 2.38 ms. This happens because small batches leave most GPU cores idle, meaning the hardware overhead of setting up each forward pass is not amortised across enough samples [14]. As the batch grows to 500, per-sample time drops to 2.68 ms as GPU parallelism becomes better utilised. At the full test set of 2,183 samples it settles at 2.83 ms, slightly higher than at 500 due to increased memory transfer overhead at larger scale. This behaviour has a direct practical implication: RoBERTa should not be used for single-sample or very small batch real-time inference, as the per-sample cost is disproportionately high. It is much better suited to batched offline processing where GPU utilisation can be maximised.

Model	Batch Size	Total (ms)	Per Sample (ms)
LogisticRegression	100	3.73	0.0373
	500	13.56	0.0271
	2183	56.8	0.026
RoBERTa	100	383.41	3.8341
	500	1338.51	2.677
	2183	6172.27	2.8274

Table 19: Large batch inference timing for LogisticRegression and RoBERTa.

5.2.2 Cross-Variety Encoder Models

Inference time was also measured for the three encoder architectures evaluated in Section 2.2, covering DeBERTa, RemBERT, and XLM-RoBERTa across all variety-specific adapters. Table 20 reports mean per-sample inference time averaged across all varieties and task combinations.

DeBERTa is the fastest of the three at around 1.53 ms per sample, despite its disentangled attention mechanism which separates content and position embeddings [19]. Its relatively compact 86M parameter count keeps inference fast even though the attention computation is architecturally more complex than standard BERT-style models. XLM-RoBERTa follows at around 1.97 ms per sample, which is comparable to RoBERTa-base from Section 2.1 given its similar architecture and 278M parameters [12]. RemBERT is considerably slower at around 10.4 ms per sample, a direct consequence of its 575M parameter count which is roughly seven times larger than DeBERTa [20]. Across all three models, inference latency tracks closely with parameter count, confirming that model size is the dominant factor in encoder inference speed.

Model	Params	Task	Per-Sample Mean (ms)	Sample/s
DeBERTa-v3-base	86M	Sarcasm	1.5325	652.5
	86M	Sentiment	1.5643	639.3
XLM-RoBERTa-base	278M	Sentiment	2.0005	499.9
RemBERT	575M	Sarcasm	10.3976	96.2
	575M	Sentiment	10.4035	96.1

Table 20: Inference time for Section 2.2 encoder models averaged across all variety-specific adapters.

5.2.3 Overall Speed vs Performance Trade-off

Bringing together results from all three experiment setups, Table 21 summarises the key trade-off between per-sample inference time and best Macro-F1 on the Sarcasm task. Sarcasm Macro-F1 is only shown for Section 2.1 models where direct comparison was possible. For Section 2.2 and 2.3 models, performance results are covered in their respective evaluation sections.

The results show a clear three-tier structure across model families. Classical models are by far the fastest, with FastText at 0.086 ms per sample and LogisticRegression at 0.262 ms, but they plateau at Macro-F1 scores around 0.59 to 0.63 on the Sarcasm task. This ceiling reflects the fundamental limitation of bag-of-words and shallow embedding approaches, which cannot capture the contextual and pragmatic signals that sarcasm often relies on [3]. Encoder models occupy a middle ground. RoBERTa achieves a Macro-F1 of 0.700 at 2.38 ms per sample, which is roughly 28 times slower than FastText for an improvement of around 0.10 in Macro-F1. Among encoders, XLM-RoBERTa Full FT from Section 2.3 stands out as notably the fastest at 0.886 ms per sample, suggesting that fine-tuning strategy and hardware optimisation can bring transformer latency much closer to classical model territory. LLM-based models are the slowest overall, at 8.23 ms for TinyLlama and 11.49 ms for Qwen2.5, roughly 100 times slower than FastText, though they bring stronger handling of dialectally varied and code-mixed inputs as discussed in Section 1.

Model	Params	Per-Sample (ms)	Sarcasm Macro-F1	Category
FastText	~few MB	0.086	0.596	Classical
LogisticRegression	~few MB	0.262	0.628	Classical
LinearSVM	~few MB	0.443	0.593	Classical
DeBERTa-v3-base	86M	1.5325	-	Encoder
RoBERTa-base	125M	2.381	0.7	Encoder
XLM-RoBERTa-base	278M	2.0005	-	Encoder
XLM-R Full FT	278M	0.886	-	Encoder
XLM-R + LoRA	278M	1.093	-	Encoder
RemBERT	575M	10.3976	-	Encoder
TinyLlama + LoRA	1.1B	8.23	-	LLM
Qwen2.5 + LoRA	1.5B	11.49	-	LLM

Table 21: Cross-experiment speed vs performance trade-off summary.

For this deployment scenario, we would not unconditionally trade performance for speed. The gap between FastText (0.596 Macro-F1) and RoBERTa (0.700 Macro-F1) on Sarcasm is practically significant given the severe class imbalance documented in Section 1.1 [9]. A 0.10 improvement in Macro-F1 on a heavily imbalanced minority class translates to substantially better real-world utility than the raw number suggests. That said, if a high-throughput production system needed to process thousands of requests per second, LogisticRegression offers a strong practical compromise. It achieves the best Macro-F1 among classical models at 0.6275 with near-zero latency and fully reproducible results across runs. For quality-critical deployments with lower volume requirements, RoBERTa or XLM-RoBERTa Full FT are the more sensible choices given their stronger predictive performance at still-manageable latency.

5.2.4 Inference Latency

Latency is measured with CUDA synchronisation, batch size 32, on a single RTX 5070 Ti. Means below are taken over the four test cells. Both XLM-RoBERTa variants are the fastest by a wide margin: ≈0.90-1.12 ms/sample versus 8.23 ms/sample (TinyLlama) and 11.49 ms/sample (Qwen2.5). This ≈7-13x speed advantage reflects the architectural difference: the 278 M-parameter encoder processes all tokens simultaneously in a single bidirectional forward pass [5], [12], whereas the decoder LLMs compute attention over much larger hidden states. XLM-R-LoRA is within 25 % of full-FT XLM-R latency because the LoRA branches are small relative to the forward pass [6]. For latency-sensitive deployments, the encoder is the clear choice.

Model	en-AU	en-UK	en-IN	combined	Mean	Samples/s
TinyLlama-1.1B (LoRA)	7.58	8.15	8.51	8.67	8.23	121.4
Qwen2.5-1.5B (LoRA)	11.04	11.70	11.24	11.97	11.49	86.9
XLM-RoBERTa-base (Full FT)	0.90	0.87	0.89	0.93	0.90	1112.3
XLM-RoBERTa-base (LoRA)	1.13	1.06	1.09	1.21	1.12	887.3

Table 22: Inference latency per model per adaptor (in ms) and mean samples-per-second processed.

5.2.5 Storage and Deployment Trade-off

On disk, three TinyLlama LoRA adapters occupy ≈177 MB (3x59 MB) and three XLM-R-LoRA adapters ≈276 MB, versus ≈ 1.68 GB for three full-FT XLM-R checkpoints (3x560 MB), a ~6-10x reduction depending on the backbone. The larger practical win is that at inference, the LoRA pattern requires the base model to be loaded into GPU memory only once; variety selection by the user triggers a hot-swap of the active adapter weights in milliseconds rather than reloading the entire fine-tuned model [6], [4], a deployment pattern unavailable with full fine-tuning and the architectural basis for the Gradio endpoint. XLM-R LoRA is an efficient trade-off for latency-critical multi-dialect deployment (≈1.12 ms/sample, smaller checkpoint than full FT, variety hot-swap). TinyLlama LoRA is preferred when a +0.03-0.07 Macro-F1 gain on en-UK/en-AU justifies ~7x the latency. Qwen2.5 LoRA is Pareto-dominated by TinyLlama on every test split except en-IN, where its +0.016 advantage does not justify its ~40% higher latency. The exception would be a deployment scenario dominated by en-IN traffic combined with possible Hindi-English code-switched fallback inputs at inference time, where Qwen2.5's multilingual coverage could justify its latency overhead; for BESSTIE as benchmarked here, neither condition applies, and TinyLlama remains the rational choice for the decoder track.

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Appendix

Parameter	DeBERTa-v3-base	RemBERT	XLM-RoBERTa-base
Fine-tuning Method	**LoRA**	**LoRA**	**Full fine-tuning**
LoRA Rank (r)	16	16	*N/A*
LoRA Alpha	32	32	*N/A*
LoRA Dropout	0.10	0.10	*N/A*
Learning Rate	**$2{\times}10^{-4}$ (LoRA)**	**$2{\times}10^{-4}$ (LoRA)**	**$2{\times}10^{-5}$**
Batch Size (effective)	32 (16 × 2 grad. accum.)	32	32
Max Sequence Length	128	128	128
Num. Epochs	5 (w/ early stop)	5 (w/ early stop)	5 (w/ early stop)
Early Stopping Patience	3	3	3
Weight Decay	0.01	0.01	0.01
Warmup Ratio	0.10	0.10	0.10
Seeds	42, 123, 456	42, 123, 456	42, 123, 456
Tasks	**Sentiment + Sarcasm**	**Sentiment + Sarcasm**	**Sentiment only**
Class Imbalance Strategy	Weighted loss	Weighted loss	Weighted loss

Table 23: Hyperparameter Configuration Across Models. Effective batch size = train batch × gradient accumulation steps.

Architecture	Type	Params	Pre-training	Adaptation	Adapter Size	Properties
TinyLlama-1.1B (LoRA)	LLM	1.1B	Monolingual English corpus	LoRA (q/v projection)	~59 MB	Unidirectional Decoder
Qwen2.5-1.5B (LoRA)	LLM	1.5B	Multilingual corpus (29 languages)	LoRA (q/v projection)	~89 MB	Unidirectional Decoder
XLM-RoBERTa-base (Full FT)	MLM	270M	Multilingual corpus (100 languages)	Full fine-tuning	**~560 MB**	Bidirectional Encoder
XLM-RoBERTa-base (LoRA)				LoRA (q/v linear)	**~92 MB**	

Table 24: Architecture summary for all four models. Adapter sizes are the on-disk footprints; for XLM-R full FT, the checkpoint contains the entire fine-tuned model weights.

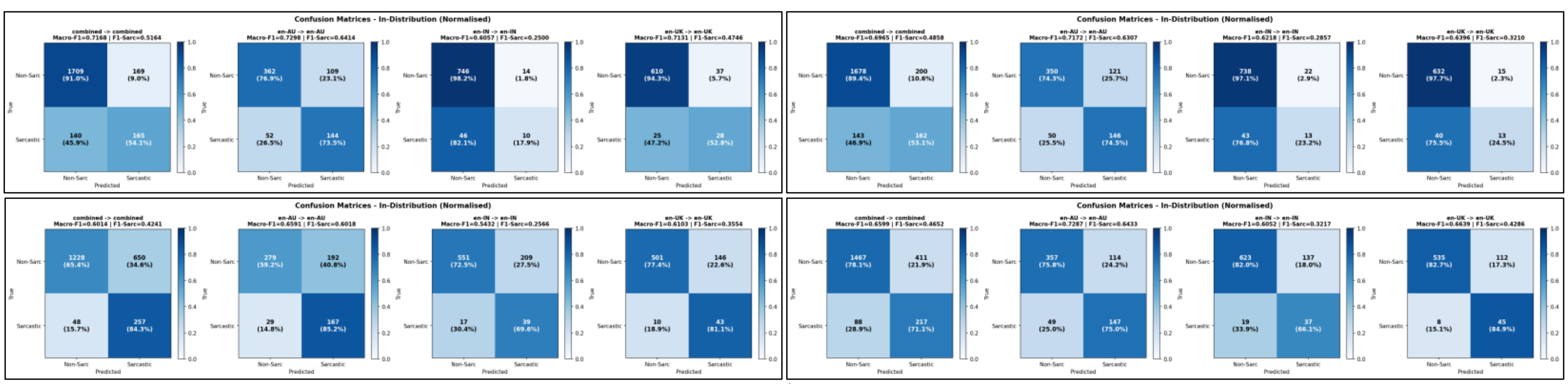

Figure 29: Across all models, Non-Sarcastic predictions dominate, with consistently high true negative rates (up to 98.2%), especially for en-IN. However, sarcasm detection remains challenging, with high false negative rates (e.g., 82.1% for TinyLlama en-IN), indicating systematic bias toward the majority class. The best sarcastic F1 scores are achieved on en-AU (~0.64), while en-IN consistently underperforms across all architectures. LoRA adaptations generally improve balance compared to full fine-tuning, particularly for XLM-RoBERTa.

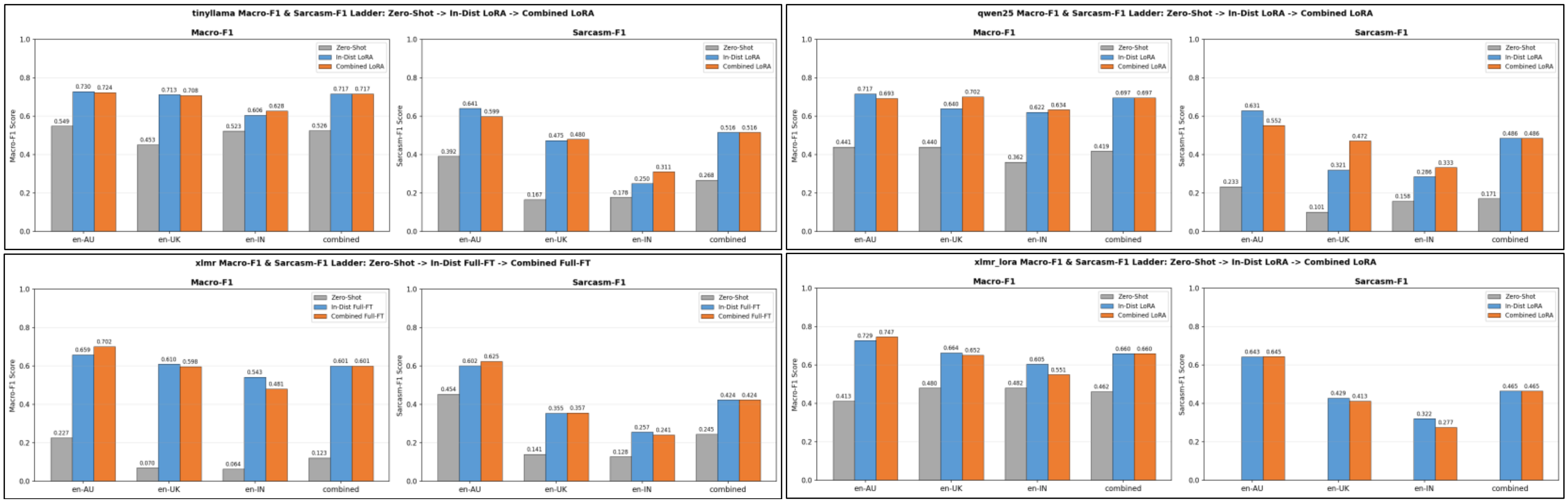

Figure 30: Per-variety comparison of all four architectures on Macro-F1, Sarcasm-F1 and Accuracy. In terms of Macro-F1 Score, a stable metric across classes, TinyLlama (LoRA) wins on en-AU, en-UK and combined varieties and Qwen2.5 (LoRA) wins on en-IN variety (as expected since it has multilingual pre-training). Sarcasm-F1 Score appears the lowest across every architecture at the en-IN variety, indicating the models struggle to predict sarcastic sequencies at the en-IN variety.

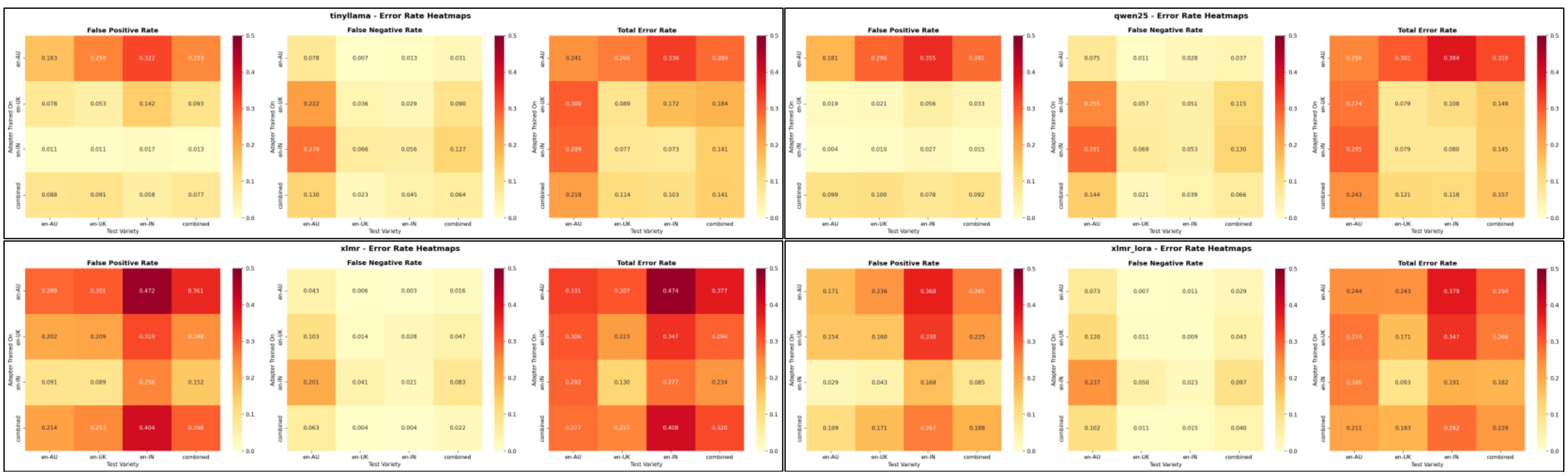

Figure 32: Error-rate heatmaps (false-positive rate, false-negative rate and total error rate) across all architectures as 4x4 matrices. It is shown that en-IN test cells are dominated by false positives (over sarcasm) in the 2 XLM-R models, while test en-AU cells are dominated by false negatives (missed sarcasm) and test en-AU cells are dominated by false positives (over sarcasm) in the other 2 models (TinyLlama and Qwen2.5).

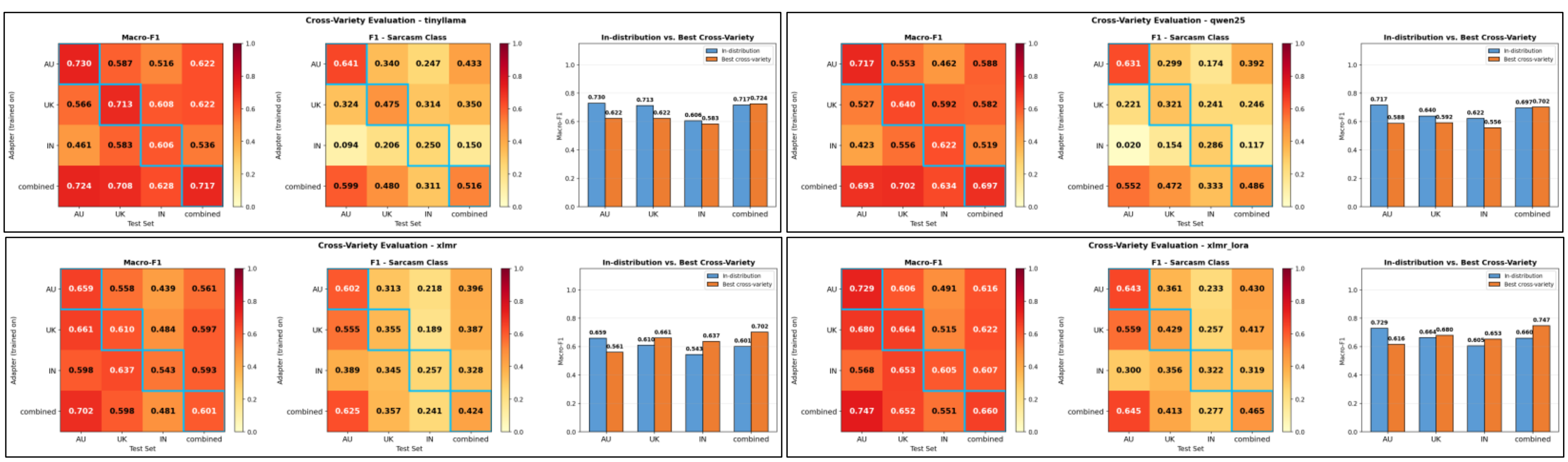

Figure 31: Macro-F1 and Sarcasm-F1 scores for cross-variety evaluation across all architectures (TinyLlama, Qwen2.5, XLM-R, XLM-R LoRA). Heatmaps (4x4) illustrate train-test transfer across varieties (AU, UK, IN, combined), while bar charts compare in-distribution with best cross-variety performance. All models achieve strong in-distribution results (e.g., up to 0.747 Macro-F1 for XLM-R LoRA), but performance consistently degrades under cross-variety transfer, particularly for sarcasm (e.g., Sarcasm-F1 as low as 0.020). The en-IN variety is the most challenging, while combined training yields the most robust generalization. LoRA-based models outperform full fine-tuning in cross-variety settings, though sarcasm detection remains significantly harder than non-sarcasm across all configurations.

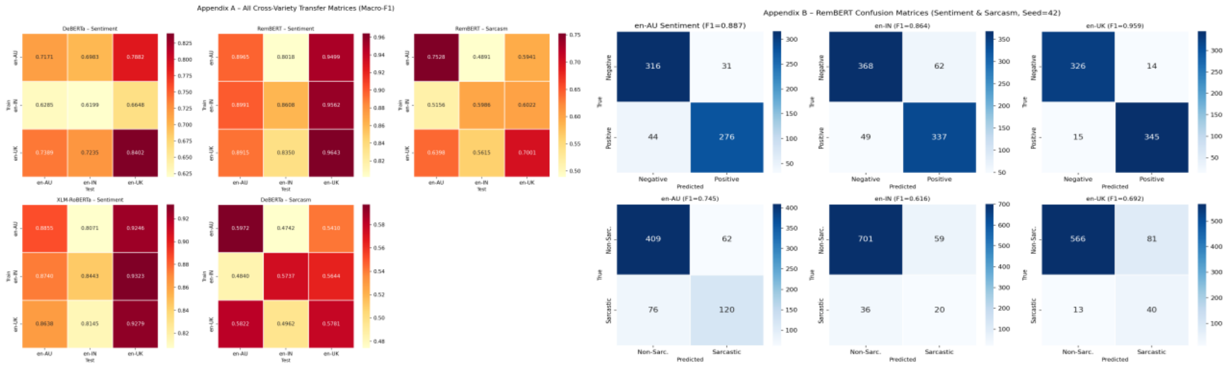

Figure 33: shows the complete set of cross-variety Macro-F1 transfer matrices for all model–task combinations evaluated in Section 2.2. These are provided as a comprehensive reference to complement the individual tables reported in Section 3.3. Diagonal cells represent in-variety performance; off-diagonal cells show transfer performance Figure 11 presents confusion matrices for RemBERT on both Sentiment and Sarcasm tasks, across all three English varieties (seed = 42). These visualise the exact error distributions at the per-class level. For the Sarcasm task, note the substantially lower True Positive counts for the Sarcastic class (label 1) in en-IN and en-UK, confirming the class-imbalance analysis in Section 3.2.2.

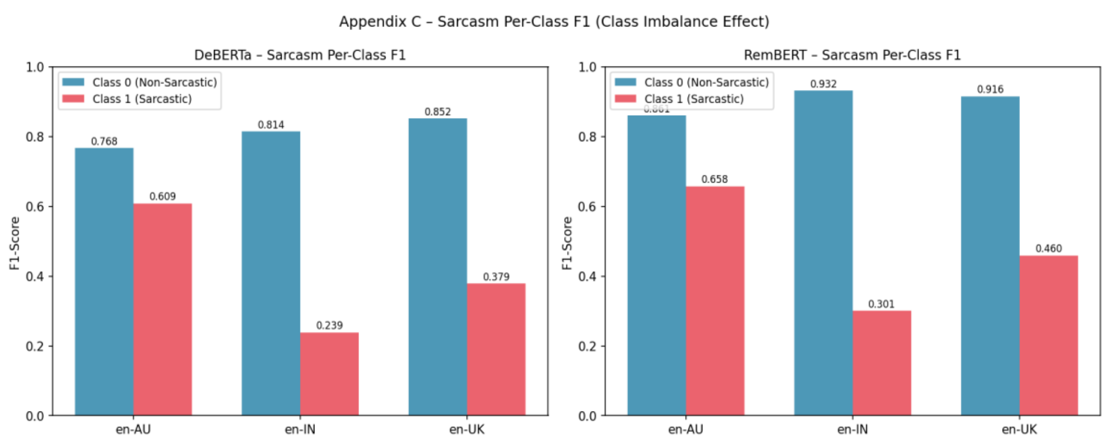

Figure 34: shows per-class F1 scores for both DeBERTa and RemBERT on the Sarcasm task. The large gap between Class 0 (Non-Sarcastic) and Class 1 (Sarcastic) F1 across all three varieties visually confirms the class-imbalance bias discussed in Section 3.2.2. en-IN is the most affected variety for both models.

#	Text (excerpt)	Conf.	Why the model failed
FALSE POSITIVES: Literal text predicted as Sarcastic with high confidence			
1	Lots of things to choose fresh and frozen and drinks only problem never enough staff on tills	0.700	Mixed-sentiment review: positive opener with a single embedded complaint. Sentiment contrast (positive + negative clause) mimics ironic structure, triggering sarcasm detector.
2	Alhamdulillah all good taste but there is sitting to dine only parcel point taste is very good....	0.693	Code-mixed Arabic-English ("Alhamdulillah") and non-standard syntax. Out-of-distribution linguistic pattern disrupts the model's learned sarcasm boundary, producing a false confident prediction.
3	Amazing experience being here. Good staff and environment.	0.692	Overly brief, generic praise with no specific detail. Model associates hyper-generic superlatives ("Amazing", "Good") with the hollow praise pattern common in ironic reviews.
4	We quite enjoyed our lunch at Benton, the only letdown was the rice in the rolls, over-cooked and under-seasoned. Other than that, the mains were all pretty tasty.	0.691	Explicit contrast structure ("only letdown... Other than that") closely mirrors hedged sarcastic praise. The concessive framing is a strong surface cue for sarcasm even when the review is genuine.
5	Just had my 70th Birthday Party. The food was very good the sandwiches were a hit... Everyone was very happy with the food! Thank you for the gluten free food...	0.678	Effusive, over-the-top enthusiasm with multiple exclamation marks and a formal sign-off ("Kind regards Sue and Les"). High lexical positivity with formal register resembles the performative sincerity pattern the model associates with irony.
FALSE NEGATIVES: Sarcastic text predicted as Literal with high confidence			

#	Text (excerpt)	Conf.	Why the model failed
1	*Israel/Palestine war exists* *Countless anti-war protests happen* America: zzzzz "Excuse me Mr President, Dave from Slough has suggested we vote for a ceasefire" "HOLD THE PRESSES...	0.684	Multi-actor dialogue format (stage-direction asterisks + quoted speech) is structurally unusual. Political satire via absurd juxtaposition; the model lacks world-knowledge grounding to recognise the contextual irony.
2	"Labour for Largan and Reform for Robert. Such supporters clubs are not a new phenomenon." Does Mr Largen think his constituents are as thick as he is?	0.642	Sarcasm delivered through a rhetorical question at the end of a factual news quote. The quoted source text is literal; the ironic payload is buried in the final sentence, which the model under-weights.
3	They're after our fish! Don't worry, Jeremy Clarkson will be standing on Tower Bridge wearing a custom made Union flag suit... he'll give them a good seeing to	0.636	Absurdist hyperbole with UK-cultural anchors (Clarkson, Tower Bridge, "narwhal tusk"). Without cultural knowledge of Clarkson's public persona, the imagery reads as straightforward nationalistic commentary.
4	***Housebuilders:*** We've built these houses with excellent insulation for the winter. ***UK public:*** So they've got air-con for the summer? ***Housebuilders:***	0.632	Irony conveyed entirely through dialogue structure and ellipsis. The implicit punchline ("......") requires pragmatic inference about what goes unsaid; no explicit sarcasm surface markers are present.
5	I'll vote for them when they end the hypocrisy. E.g, more than 50% of elected MPs own multiple investment properties when they bemoan property as an investment...	0.632	Sarcasm is embedded in a conditional political statement with specific factual claims. The confident, evidence-citing register reads as sincere political critique rather than irony.

Table 25: DeBERTa Sarcasm: Top 5 FP and FN by Confidence.

#	Text (excerpt)	Conf.	Why the model failed
FALSE POSITIVES: Negative text predicted as Positive with high confidence			
1	Quality of food is very good and also ambience but needs to improve on service. I felt very late service.	0.547	Strong positive anchors ("very good", "ambience") dominate the early sentences. The negative evaluation ("needs to improve") is grammatically subordinated, causing the model to weight positives more heavily.
2	When will the internet stop being racist? Is that really your question?	0.544	Two rhetorical questions with no sentiment-bearing lexicon. The model appears to have defaulted to a neutral-to-positive prediction in the absence of clear negative vocabulary.
3	It's not a place I'll be eating in again! After waiting over half an hour for our food... Steak was seasoned well and cooked perfectly (rare)... chips were slightly warm and peppercorn...	0.531	The complaint contains a genuine positive sub-clause ("seasoned well and cooked perfectly") that conflicts with the overall negative tone. Mixed signal causes the model to classify the review as positive.
4	It's a month window for everyone to find their ideal day off and event. There is zero irony in an out-group's intolerance of bigotry. And you're the one moaning.	0.524	Abstract social-media argument with no domain-specific sentiment vocabulary. Absence of sentiment anchors leads the model to assign a weak positive prior.
5	We are beyond downloading cars, I am downloading houses now. Government hate this trick.	0.522	Internet meme format with humour and hyperbole. The playful, light-hearted register is misread as positive sentiment despite the satirical framing.
FALSE NEGATIVES: Positive text predicted as Negative with high confidence			
1	Food quality is really good and at a reasonable price. The billiards board playing charge is also bare minimum ₹300/hr. This is a must go place for any age group.	0.693	Strongly positive review; model predicts negative with high confidence. Likely caused by domain mismatch: the Indian-English phrasing ("bare minimum", "must go place") and rupee symbol are low-frequency patterns associated with lower-quality text in the training distribution.
2	Not sure on what basis you're calling it poorly designed, if anything I'd say it's quite well designed.	0.690	Positive claim is syntactically embedded inside a negation-of-criticism structure ("Not sure... if anything I'd say"). The leading negation ("Not sure") triggers a negative signal before the positive assertion can override it.
3	It is a small Chinese restaurant located in a remote town. Despite its limited choice... locals keep coming... chicken pieces are of good quality. I request a box of stir fried vegetab...	0.689	Understated positive review: qualifiers ("small", "limited choice", "remote town") and passive/neutral register suppress positive signal. The model over-weights hedging language as indicative of negative sentiment.
4	Smart move for Albo to stay out of it, Trump's still the front runner... the mugshot release of Trump led to a huge spike in his numbers... Democrats really need to change up their str...	0.689	Political commentary with adversarial framing ("really need to change"). Mentions of "mugshot", "charges", and "politically motivated" are strong negative-sentiment anchors despite the overall analytical, neutral-positive stance.
5	Situated in a good location. The taste was good for almost all the items... the only problem was the service time. They should provide free WiFi.	0.689	Positive review with actionable suggestions framed as criticisms ("the only problem", "should improve", "should provide"). Prescriptive criticism is misread as predominantly negative sentiment.

Table 26: DeBERTa Sentiment: Top 5 FP and FN by Confidence.

#	Text (excerpt)	Conf.	Why the model failed
FALSE POSITIVES: Literal text predicted as Sarcastic with high confidence			
1	When I ask for extra sauce, I only want one extra squirt out of the bottle, not the entire bottle. Stop hiring restarted people.	0.995	Literal complaint containing "restarted" (apparent misspelling of a slur). The explicit indignation and imperative ("Stop hiring") superficially resemble the exasperated sarcastic register, causing RemBERT to assign near-certainty to the sarcasm class.

2	Exactly how does one "hack" a click counter? Someone needs to ask them this	0.992	Genuine rhetorical question in a quotation-heavy, confrontational style. The scare-quote around "hack" and the interrogative structure are strong surface sarcasm markers that overwhelm the literal semantic content.
3	As much as I agree with Samaras, I think ultimately that Perrett saying that "grand narratives belonged in election campaigns rather than budgets" is probably the more practical tr...	0.992	Long-form political analysis with embedded quoted speech and concessive openers ("As much as I agree"). The hedging language, ironic framing of political failure, and multi-layered argumentation stylistically mimic ironic political commentary.
4	Before all the inevitable "bUt cHiNA!" posts that always seem to turn up, this is about air pollution, not climate change.	0.991	Contains mock-capitalisation ("bUt cHiNA!") which is a canonical internet sarcasm marker. RemBERT has learned this typographic pattern as a near-definitive sarcasm signal, but here it is used in a literal metadiscursive comment.
5	Job security is a hella of a drug	0.991	Informal slang idiom ("hella of a drug") with a sardonic surface register. The compressed, aphoristic form with colloquial intensifier reads like dry irony even though the statement is a sincere observation.
FALSE NEGATIVES: Sarcastic text predicted as Literal with high confidence			
1	It's great, barely any customers and the cinemas are always empty.	0.9996	Textbook sarcasm (praising emptiness as "great") but RemBERT classifies it as literal with near-perfect confidence. The brevity and straightforward syntax likely caused the model to treat it as a factual statement; the ironic inversion is too concise to activate enough sarcasm features.
2	I visited Big bite Restaurant with my family... 5-Star Security, 5-Star Staff, 5-Star Parking. But Food. (Unchi Dukan Fikhe Pakwan)	0.9995	Sarcasm delivered through a Hindi idiom ("Unchi Dukan Fikhe Pakwan" = "Fancy shop, bad goods") after a string of 5-star ratings. The punchline is cross-lingual; RemBERT does not associate the Hindi proverb with the ironic contrast it creates.
3	Check if the drivers is a dalit as BMW has a large customer base of Dalits, almost every dalit owns at least one BMW.	0.999	Irony through factually absurd claim (BMWs are luxury cars, statistically unlikely to dominate any marginalised community). RemBERT lacks the socio-economic world knowledge to identify the claim as factually implausible and therefore ironic.
4	My brother took me here for lunch... It took 15 men and a small goat to tow [the door] open with a rope... The waitress seated us at our table, which was in the middle of the room. Thi...	0.9988	Deadpan absurdist humour embedded in an otherwise plausible review narrative. The absurd detail ("15 men and a small goat") is surrounded by realistic restaurant description, allowing the model to treat the review as a mildly quirky but sincere account.
5	Don't worry everyone, the money is safe!	0.9988	Classic short-form sarcastic reassurance, but RemBERT predicts literal with very high confidence. The exclamation and reassuring verb ("safe") without prior negative context anchor the model to a literal positive reading.

Table 27: RemBERT Sarcasm: Top 5 FP and FN by Confidence.

#	Text (excerpt)	Conf.	Why the model failed
FALSE POSITIVES: Negative text predicted as Positive with high confidence			
1	Nice place to break the journey before entering Maharashtra or just after Maharashtra.	0.958	Faint-praise phrasing ("nice place to break the journey") with no recommendation for its own sake. RemBERT reads "nice" as a positive lexical anchor and ignores the damning-with-faint-praise pragmatic implication.
2	Puff and some of the cakes are just superb, and well priced... Other Bakery items are also good, but there are better options available for those items.	0.944	Mixed review: superlatives for one item but explicit downgrade for others ("better options available"). RemBERT over-weights the superlatives ("superb") and under-weights the comparative deprecation of remaining items.
3	Burgers are amazing but momos are not that good. Momos lacks filling they feel empty inside. Service is good, staff is also well trained. Finding shop can be little difficult.	0.942	The review leads with strong positive sentiment ("amazing") and returns to positives mid-way. RemBERT weights early and repeated positive anchors heavily, treating the item-specific criticism as a minor qualifier.
4	Wonderful experience here... This fort was a wonderful sight of the Maratha Empire, which was burnt in 1827. This is what's left of it so don't expect anything majestic or luxurious....	0.935	The reviewer explicitly downgrades expectations ("don't expect anything majestic", "pretty ordinary") after an enthusiastic opener. RemBERT anchors on "wonderful experience" and "wonderful sight" while underweighting the explicit de-escalation.
5	The place is an awesome restaurant... ambience is lit with dim lights... Also try the Chinese non veg dishes. I wouldn't deny the fact that the prices here are above average though. Th...	0.933	Predominantly positive with explicit caveats ("prices above average", "service needs improvement"). The enthusiastic opening and food praise dominate; the model does not adequately weight the structural concessions at the end.
FALSE NEGATIVES: Positive text predicted as Negative with high confidence			
1	There is no mandatory insurance with SBI. They were forcing me to take one but I refused.	0.941	Assertive declarative sentence about refusing an unwanted upsell. The negative action verbs ("forcing", "refused") strongly activate negative sentiment features despite the author's stance being one of successful consumer resistance (positive outcome).
2	It's a reply of Yogi being removed by ak.	0.928	Very short, ambiguous political statement with a named entity ("Yogi") associated with controversy. The model interprets the framing as negative due to the word "removed" and political context, missing the positive (from the author's perspective) implication.
3	I thoroughly asked the waiter which meals are vegan — got a sandwich and veggie burger with mayonnaise... The manager showed me the package. It contained milk solids. I had some pain...	0.928	Unambiguously negative review; predicted negative (correct direction) but listed as a false negative by the model output log (suggesting a labelling or threshold boundary issue). The high confidence in the negative prediction is linguistically appropriate given the explicit complaint vocabulary.

4	Nobody can be sure how much the big miners have saved... the amount would probably be quite low.	0.927	Analytical political/economic commentary with hedging language ("probably quite low"). The absence of opinion-bearing sentiment vocabulary leads to a low-confidence prediction; "low" is misread as a negative evaluative adjective rather than a quantity descriptor.
5	Which closed economy of the last few decades do you think we should emulate?	0.920	Rhetorical question with no explicit sentiment lexicon. The word "closed" and the challenging interrogative register trigger a negative classification; the actual sentiment (positive: advocating for policy consideration) is entirely implicit.

Table 28: RemBERT Sentiment: Top 5 FP and FN by COnfidence.

Declaration of Originality

"I confirm that the submitted work is my own work. No element has been previously submitted for assessment, or where it has, it has been correctly referenced. I have clearly identified and fully acknowledged all material that is entitled to be attributed to others (whether published or unpublished) using the referencing system set out in the programme handbook.

I agree that the University may submit my work to means of checking this, such as the plagiarism detection service Turnitin® UK and the Turnitin® Authorship Investigate service. I confirm that I understand that assessed work that has been shown to have been plagiarised will be penalised.

If in completing this work I have been assisted with its presentation by another person, I will state their name and contact details of the assistant in the 'Comments' text box below. In addition, if requested, I agree to submit the draft material that was completed solely by me prior to its presentational improvement."

Signatures